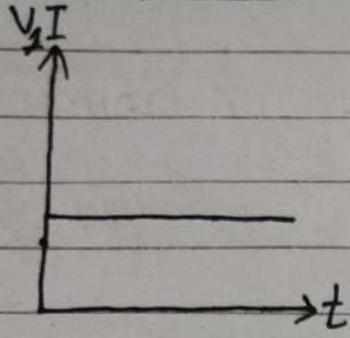


Module - I

1. Define Circuit (ckt)
2. Mesh
3. Loop
4. Node
5. Junction
6. Branch
7. Active Element
8. Passive Element
9. Linear Network (N/W)
10. Non-linear Network
11. Difference between DC & AC
12. Circuit Parameters
13. Sources
14. KVL
15. KCL
- *16. Nodal Analysis
17. Mesh Analysis
18. Star-to-Delta Transformation
19. Delta-to-Star Transformation
20. Source Transformation
21. Voltage-to-Current Transformation
22. Theorem
 - * (a) Thevenin's Theorem
 - (b) Maximum Power Transfer Theorem
 - (c) Norton's Theorem
 - (d) Superposition Theorem
 - (e) Millman's Theorem
 - (f) Tellegen's Theorem.

Difference Between DC & AC

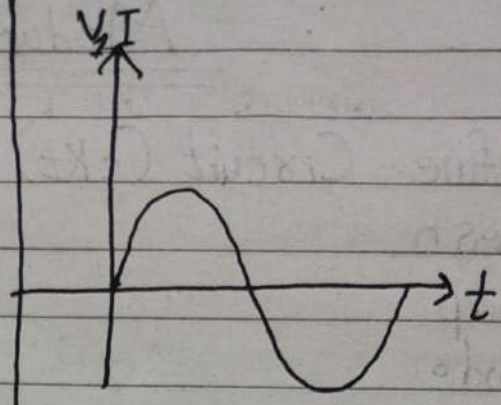
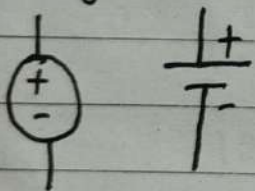


DC

Voltage & Current constant with respect to the time.

Invariant System.

ckt symbol;

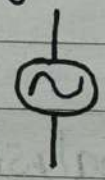


AC

Voltage & Current aren't constant.

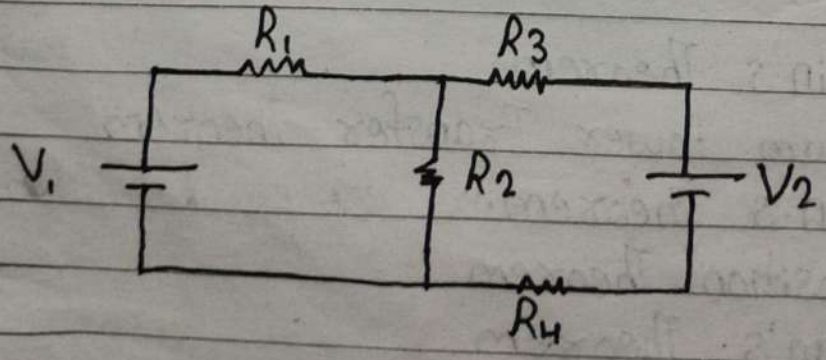
Variant System.

ckt symbol;



Circuit (ckt)

Closed conducting path is called a circuit.



Active Element

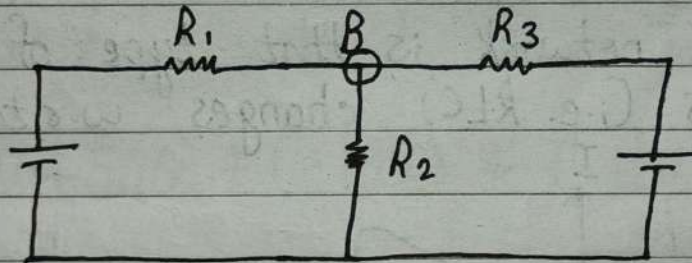
A circuit or network which contains the energy sources is called an active element.

Passive Element

A circuit or network which doesn't contain energy sources is called a passive element.

Node

Two or more branches meeting at any one point is called a node. Denoted by $V_1, \dots, V_2, V_3, \dots$.



Junction

Three or more branches meet at any one point is called a junction.

Mesh

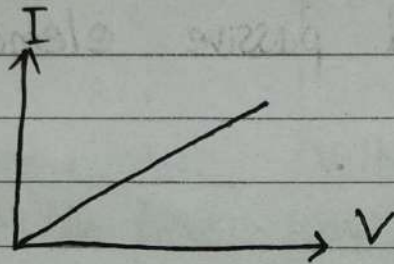
Basic closed path that cannot be divided is called a mesh.

Loop

All closed path is called a loop.

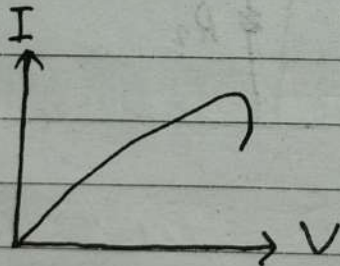
Linear Network

It is that type of network whose parameters (i.e. RLC) are constant.



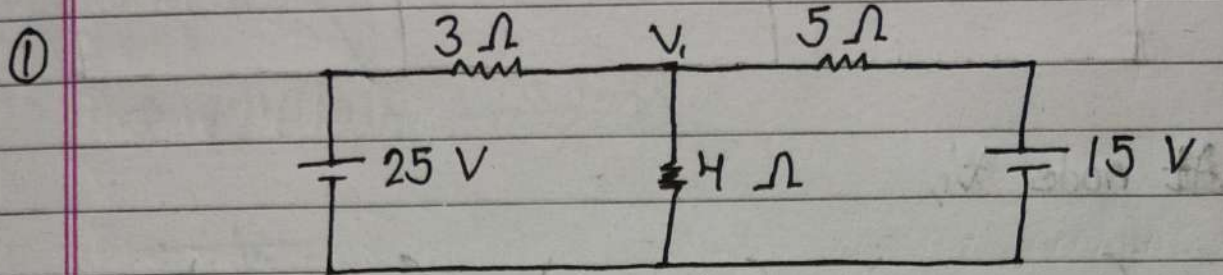
Non-linear Network

Non-linear network is that type of network whose parameters (i.e. RLC) changes w.r.to the time.



* Nodal Analysis (KCL)

Technique to find node voltage and current.



From KCL,
 $\sum I = 0$

At node V_1 ,

$$\left(\frac{-V_1 + 25}{3} \right) + \left(\frac{-V_1 + 0}{4} \right) + \left(\frac{-V_1 + 15}{5} \right) = 0 \quad [\because I = V/R]$$

$$\Rightarrow (-20V_1 + 500) + (-15V_1) + (-12V_1 + 180) = 0$$

$$\Rightarrow -47V_1 + 680 = 0$$

$$\Rightarrow 47V_1 = 680$$

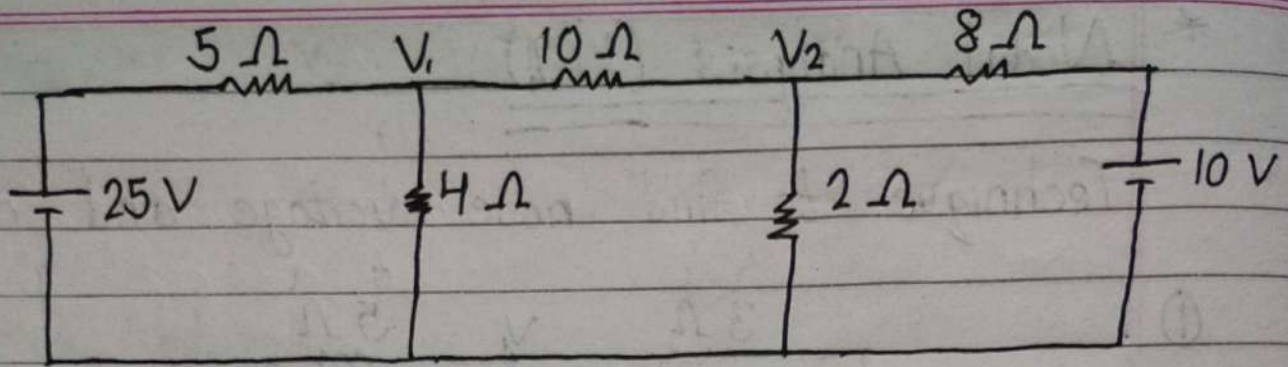
$$\therefore V_1 = \frac{680}{47} = 14.46 \text{ V}$$

$$\text{Also, } I_{3\Omega} = \frac{-14.46 + 25}{3} = 3.513 \text{ A}$$

$$I_{4\Omega} = -3.615 \text{ A}$$

$$I_{5\Omega} = 0.108 \text{ A} \quad \#$$

②

At node V_1 ,

$$\left(\frac{-V_1 + 25}{5} \right) + \left(\frac{-V_1 + 0}{4} \right) + \left(\frac{-V_1 + V_2}{10} \right) = 0 \quad \text{--- ①}$$

At node V_2 ,

$$\left(\frac{10 - V_2}{8} \right) + \left(\frac{-V_2}{2} \right) + \left(\frac{-V_2 + V_1}{10} \right) = 0 \quad \text{--- ②}$$

From ①,

$$(-40V_1 + 1000) + (-50V_1) + (-20V_1 + 20V_2) = 0$$

$$\Rightarrow -110V_1 + 1000 + 20V_2 = 0$$

$$\Rightarrow V_2 = \frac{110V_1 - 1000}{20} = \frac{11V_1 - 100}{2}$$

From ②,

$$(200 - 20V_2) + (-80V_2) + (-16V_2 + 16V_1) = 0$$

$$\Rightarrow 200 - 116V_2 + 16V_1 = 0$$

$$\Rightarrow 16V_1 = 116 \left(\frac{110V_1 - 1000}{20} \right) - 200$$

$$\Rightarrow 320V_1 = 12760V_1 - 116000 - 23200$$

$$\Rightarrow 12440V_1 = -139200$$

$$\therefore V_1 =$$

$$200 - 116V_2 + 16V_1 = 0$$

$$\Rightarrow 50 - 29V_2 + 4V_1 = 0$$

$$\Rightarrow 50 + 4V_1 = 29V_2$$

$$\Rightarrow 50 + 4V_1 = 29 \left(\frac{11V_1 - 100}{2} \right)$$

$$\Rightarrow 100 + 8V_1 = 319V_1 - 2900$$

$$\Rightarrow 3000 = 311V_1$$

$$\therefore V_1 = 9.64 \text{ V}$$

$$\text{And, } V_2 = \frac{11 \times 9.64 - 100}{2} = 3.015 \text{ V}$$

$$I_{5\Omega} = 3.072 \text{ A}$$

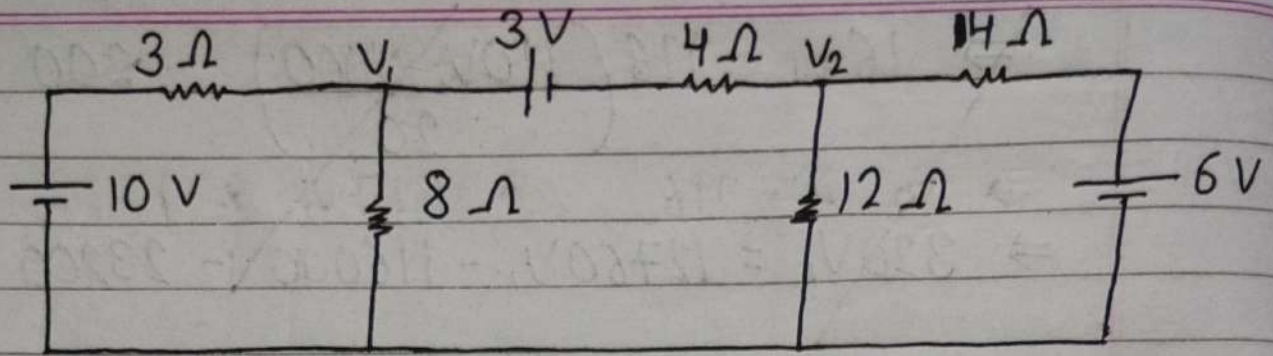
$$I_{2\Omega} = -1.525 \text{ A}$$

$$I_{4\Omega} = -2.41 \text{ A}$$

$$I_{8\Omega} = 0.8687 \text{ A}$$

$$I_{10\Omega} = -0.659 \text{ A}$$

③

At node V_1 ,

$$\left(\frac{-V_1 + 10}{3} \right) + \left(\frac{-V_1}{8} \right) + \left(\frac{-V_1 + V_2 + 3}{4} \right) = 0 \quad \text{--- ①}$$

At node V_2 ,

$$\left(\frac{6 - V_2}{14} \right) + \left(\frac{-V_2}{12} \right) + \left(\frac{-V_2 + V_1 + 3}{4} \right) = 0 \quad \text{--- ②}$$

From ①,

$$(-32V_1 + 320) + (-12V_1) + (-24V_1 + 24V_2 + 72) = 0$$

$$\Rightarrow -68V_1 + 392 + 24V_2 = 0$$

$$\Rightarrow 4(-17V_1 + 98 + 6V_2) = 0$$

$$\Rightarrow V_2 = \frac{17V_1 - 98}{6}$$

From ②,

$$(288 - 48V_2) + (-56V_2) + (-168V_2 + 168V_1 - 504) = 0$$

$$\Rightarrow -216 - 272V_2 + 168V_1 = 0$$

$$\Rightarrow 168V_1 - 216 = \frac{136}{272} \left(\frac{17V_1 - 98}{63} \right)$$

$$\Rightarrow 4(42V_1 - 54) = \frac{34}{136} \left(\frac{17V_1 - 98}{3} \right)$$

$$\Rightarrow 126V_1 - 162 = 578V_1 - 3332$$

$$\Rightarrow 3170 = 452V_1$$

$$\therefore V_1 = 7.013 \text{ V}$$

$$\text{And, } V_2 = \frac{17 \times 7.013 - 98}{6} = 3.536 \text{ V}$$

$$I_{3\Omega} = \frac{-7.013 + 10}{3} = 0.9956 \text{ A}$$

$$I_{8\Omega} = -0.876 \text{ A}$$

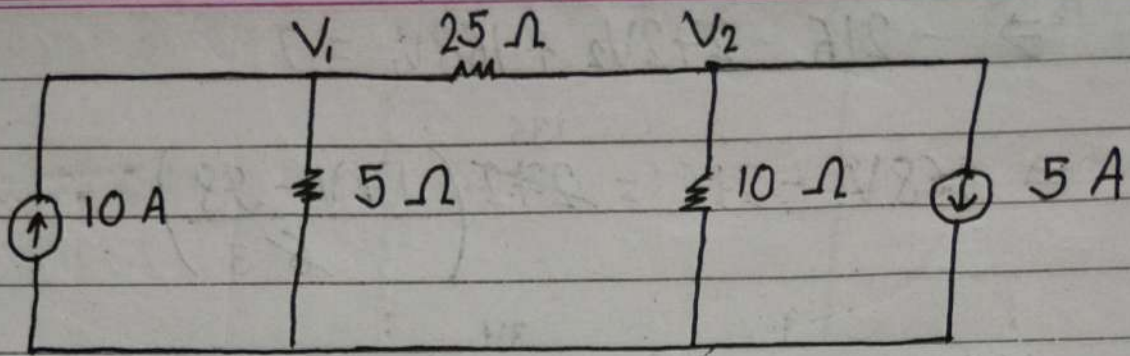
$$I_{4\Omega} = -0.11925 \text{ A}$$

$$I_{14\Omega} = 0.176 \text{ A}$$

$$I_{12\Omega} = -0.2946 \text{ A}$$

$$I_{4\Omega} = 0.11925 \text{ A} \quad \#$$

④



At node V_1 ,

$$\left(\frac{-V_1 + 0}{5} \right) + \left(\frac{-V_1 + V_2}{25} \right) + 10 = 0 \quad \text{--- (1)}$$

At node V_2 ,

$$\left(\frac{-V_2 + 0}{10} \right) + \left(\frac{-V_2 + V_1}{25} \right) - 5 = 0 \quad \text{--- (2)}$$

From (1),

$$(-25V_1) + (-5V_1 + 5V_2) + 1250 = 0$$

$$\Rightarrow -30V_1 + 5V_2 + 1250 = 0$$

$$\Rightarrow -6V_1 + V_2 + 250 = 0$$

$$\Rightarrow V_1 = \frac{V_2 + 250}{6}$$

From ②,

$$-25V_2 + (-10V_2 + 10V_1) - 1250 = 0$$

$$\Rightarrow -35V_2 + 10V_1 - 1250 = 0$$

$$\Rightarrow -35V_2 + 10\left(\frac{V_2 + 250}{6}\right) = 1250$$

$$\Rightarrow -35V_2 + 1.66V_2 + 416.66 = 1250$$

$$\Rightarrow -33.34V_2 = 833.34$$

$$\therefore V_2 = -24.99 \text{ V}$$

And,

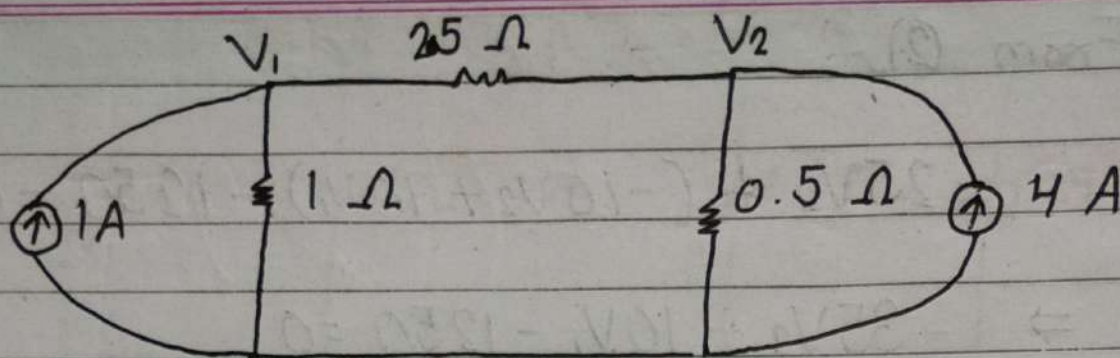
$$V_1 = \frac{-24.99 + 250}{6} = 37.5016 \text{ V}$$

$$I_{5\Omega} = \frac{-37.5016}{5} = -7.5 \text{ A}$$

$$I_{25\Omega} = \frac{37.5016 + 24.99}{25} = 2.49 \text{ A}$$

$$I_{10\Omega} = \frac{24.99}{10} = 2.499 \text{ A}$$

⑤



At node V_1 ,

$$\left(\frac{-V_1 + 0}{1} \right) + \left(\frac{-V_1 + V_2}{2.5} \right) + 1 = 0 \quad \text{--- ①}$$

At node V_2 ,

$$\left(\frac{-V_2 + 0}{0.5} \right) + \left(\frac{-V_2 + V_1}{2.5} \right) + 4 = 0 \quad \text{--- ②}$$

From ①,

$$\begin{aligned} -2.5V_1 - V_1 + V_2 &= -1 \times 2.5 \\ \Rightarrow -3.5V_1 + V_2 &= -2.5 \\ \Rightarrow V_2 &= 3.5V_1 - 2.5 \end{aligned}$$

$$\Rightarrow \underline{\underline{-77.5 V_1}}$$

From ②,

$$-2.5V_2 - 0.5V_2 + 0.5V_1 + 5 = 0$$

$$\text{or, } -3V_2 + 0.5V_1 + 5 = 0$$

$$\text{or, } -3(2.5V_1 - 2.5) + 0.5V_1 + 5 = 0$$

$$\text{or, } \underline{\underline{-78V_1}} + 7.5 + 0.5V_1 + 5 = 0 + 10.5V_1$$

$$\text{or, } \underline{\underline{-77.5V_1}}$$

$$\text{or, } 10V_1 = 12.5$$

$$\therefore V_1 = 1.25 \text{ V}$$

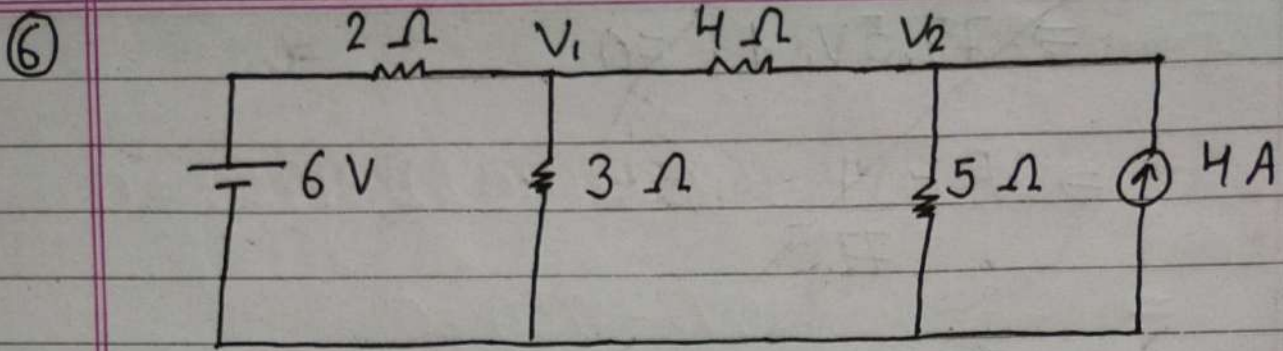
And,

$$V_2 = 3.5 \times 1.25 - 2.5 = 1.875 \text{ V}$$

$$I_{1\Omega} = -1.25 \text{ A}$$

$$I_{2.5\Omega} = \frac{-1.25}{2.5} + \frac{1.875}{2.5} = 0.625 \text{ A} = 0.25 \text{ A}$$

$$I_{0.5\Omega} = \frac{-1.875}{0.5} = -3.75 \text{ A} \neq$$



At node V_1 ,

$$\left(\frac{-V_1}{3} \right) + \left(\frac{-V_1 + 6}{2} \right) + \left(\frac{-V_1 + V_2}{4} \right) = 0 \quad \text{--- ①}$$

At node V_2 ,

$$\left(\frac{-V_2}{5} \right) + \left(\frac{-V_2 + V_1}{4} \right) + 4 = 0 \quad \text{--- ②}$$

From ①,

$$\cancel{-8V_1} + \cancel{(-3V_1)}$$

$$-8V_1 - 12V_1 + 72 + (-6V_1 + 6V_2) = 0$$

$$\text{or, } -26V_1 + 72 + 6V_2 = 0$$

$$\text{or, } V_2 = \frac{26V_1 - 72}{6} = \frac{13V_1 - 36}{3}$$

From ②,

$$-4V_2 - 5V_2 + 5V_1 + 80 = 0$$

$$\text{or, } -9V_2 + 5V_1 + 80 = 0$$

$$\text{or, } \cancel{-9} \left(\frac{13V_1 - 36}{\cancel{3}} \right) + 5V_1 + 80 = 0$$

$$\begin{aligned} \text{or, } -39V_1 + 108 + 5V_1 + 80 &= 0 \\ \text{or, } -34V_1 &= -188 \\ \therefore V_1 &= 5.529 \text{ V} \end{aligned}$$

And,

$$V_2 = \frac{13 \times 5.529 - 36}{3} = 11.959 \text{ V}$$

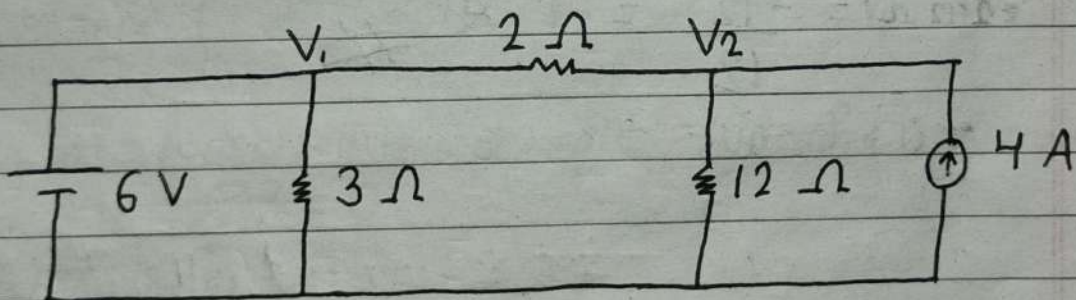
$$I_{2\Omega} = \frac{-5.529 + 6}{2} = 0.235 \text{ A}$$

$$I_{3\Omega} = -1.843 \text{ A}$$

$$I_{4\Omega} = 1.6075 \text{ A}$$

$$I_{5\Omega} = -2.39 \text{ A} \neq$$

(7)



If no resistance is present between source voltage and node then voltage at point node will be equal to the source voltage.

$$\text{i.e. } V_1 = 6 \text{ V} \quad \dots \text{ (1)}$$

At node V_2 ,

$$\left(\frac{-V_2}{12} \right) + \left(\frac{-V_2 + V_1}{2} \right) + 4 = 0$$

$$\text{or, } -2V_2 - 12V_2 + 12V_1 + 96 = 0$$

$$\text{or, } -14V_2 + 12 \times (6) + 96 = 0$$

$$\text{or, } -14V_2 = -168$$

$$\therefore V_2 = 12 \text{ V}$$

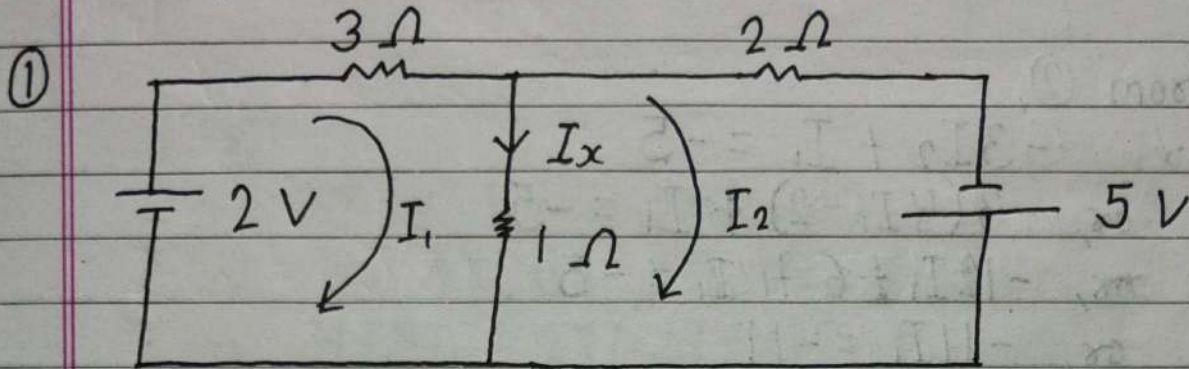
$$I_{3 \Omega} = \frac{-6}{3} = -2 \text{ A}$$

$$I_{2 \Omega} = \frac{-6 + 12}{2} = 3 \text{ A}$$

$$I_{12 \Omega} = \frac{-12}{12} = -1 \text{ A} \quad \#$$

Mesh Analysis (KVL)

Technique to find unknown currents.



Steps:

- (i) Find mesh
- (ii) Current direction (Clock-wise direction)
- (iii) $- \rightarrow +ve$ (+ve)
 $+ \rightarrow -ve$ (-ve)
- (iv) Resistance (-ve sign)

No. of mesh = No. of eqⁿ. = No. of unknown
 $\sum V = 0$

At mesh I_1 ,

$$2 - 3I_1 - 1(I_1 - I_2) = 0 \quad \text{--- ①}$$

~~or, $-4I_1$~~

At mesh I_2 ,

$$5 - 1(I_2 - I_1) - 2I_2 = 0 \quad \text{--- ②}$$

From ①,

$$-4I_1 + I_2 = -2$$

$$\text{or, } I_2 = 4I_1 - 2$$

From ②,

$$-3I_2 + I_1 = -5$$

$$\text{or, } -3(4I_1 - 2) + I_1 = -5$$

$$\text{or, } -12I_1 + 6 + I_1 = -5$$

$$\text{or, } -11I_1 = -11$$

$$\therefore I_1 = 1 \text{ A}$$

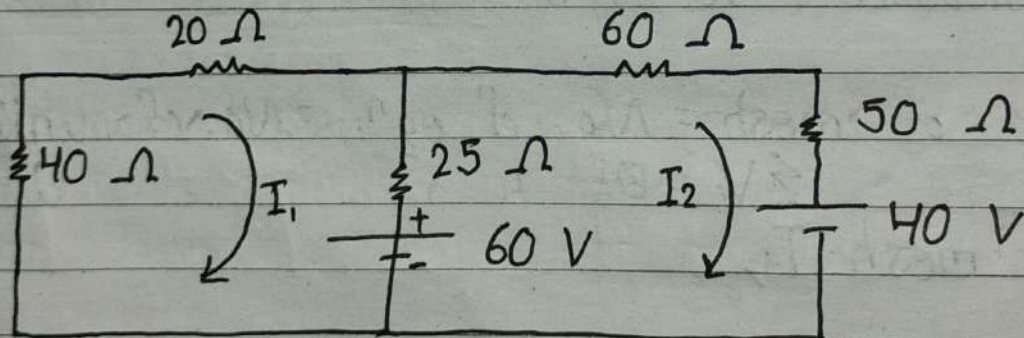
And,

$$I_2 = 4 \times 1 - 2 = 2 \text{ A}$$

$$\text{Therefore, } I_x = I_1 - I_2 = -1 \text{ A}$$

$$\& I_x = I_2 - I_1 = 1 \text{ A} \quad \#$$

②



For mesh I_1 ,

$$-40I_1 - 20I_1 - 25(I_1 - I_2) - 60 = 0 \quad \text{--- ①}$$

For mesh I_2 ,

$$+60 - 25(I_2 - I_1) - 60I_2 - 50I_2 - 40 = 0 \quad \text{--- ②}$$

From ①,

$$-85I_1 + 25I_2 = 60$$

$$\text{or, } I_2 = \frac{60 + 85I_1}{25} = \frac{12 + 17I_1}{5}$$

From ②,

$$20 - 135I_2 + 25I_1 = 0$$

$$\text{or, } 4 - 27I_2 + 5I_1 = 0$$

$$\text{or, } 4 + 5I_1 = 27 \left(\frac{12 + 17I_1}{5} \right)$$

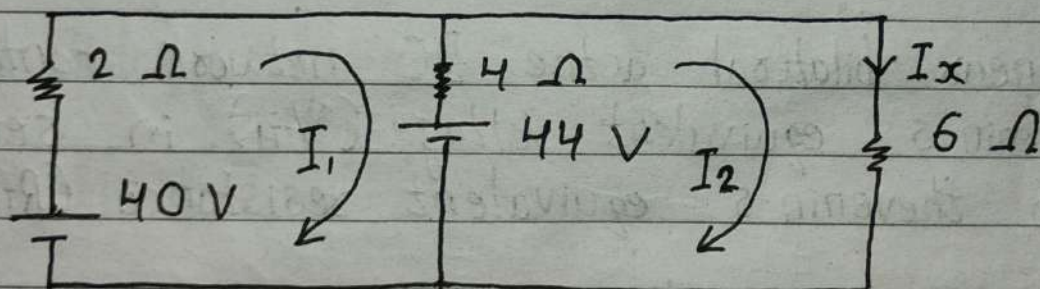
$$\text{or, } 20 + 25I_1 = 324 + 459I_1$$

$$\text{or, } -304 = 434I_1$$

$$\therefore I_1 = -0.7 \text{ A}$$

And,

$$I_2 = \frac{12 - (0.7 \times 17)}{5} = 0.02 \text{ A} \quad \#$$



For mesh I_1 ,

$$40 - 2I_1 - 44 - 4(I_1 - I_2) = 0 \quad \text{--- ①}$$

For mesh I_2 ,

$$44 - 4(I_2 - I_1) - 6I_2 = 0 \quad \text{--- ②}$$

From ①,

$$-4 - 6I_1 + 4I_2 = 0$$

$$\therefore, I_2 = \frac{6I_1 + 4}{4} = \frac{3I_1 + 2}{2}$$

From ②,

$$44 - 10I_2 + 4I_1 = 0$$

$$\therefore, 44 + 4I_1 = \frac{10(3I_1 + 2)}{2}$$

$$\therefore, 44 + 4I_1 = 15I_1 + 10$$

$$\therefore, 34 = 11I_1$$

$$\therefore I_1 = 3.09 \text{ A}$$

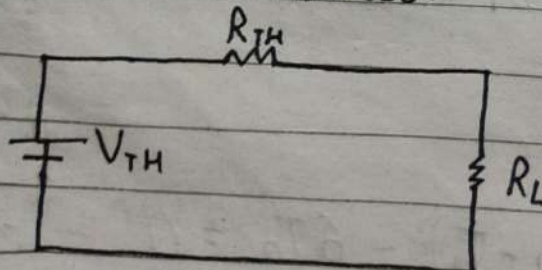
&

$$I_2 = \frac{3 \times 3.09 + 2}{2} = 5.635 \text{ A} \#$$

Imp:Thevenin's Theorem

"A linear bilateral active DC network containing thevenin's equivalent voltage (V_{TH}) in series with thevenin's equivalent resistance (R_{TH}).

Where, V_{TH} is open-circuit voltage across the load resistance and R_{TH} , when energy source (i.e. voltage & current source) are replaced to their internal resistances."

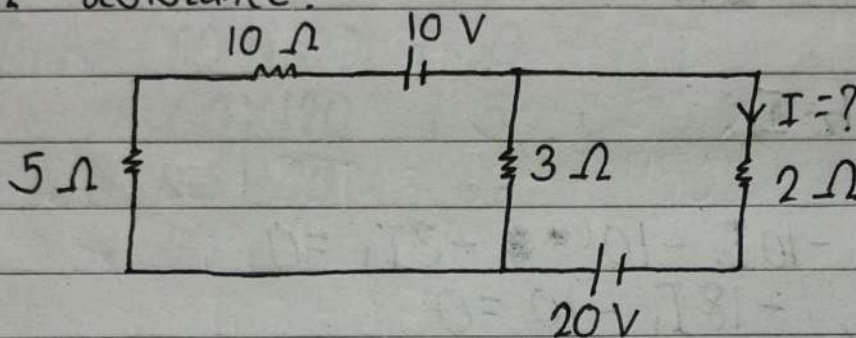


Thevenin's equivalent ckt,

$$I = \frac{V_{TH}}{R_{TH} + R_L}$$

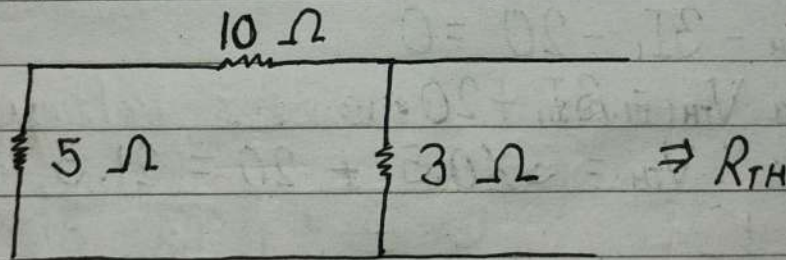
It is applicable only for linear network.

- (Q) Using Thevenin's Theorem find out the current in $2\ \Omega$ resistance?



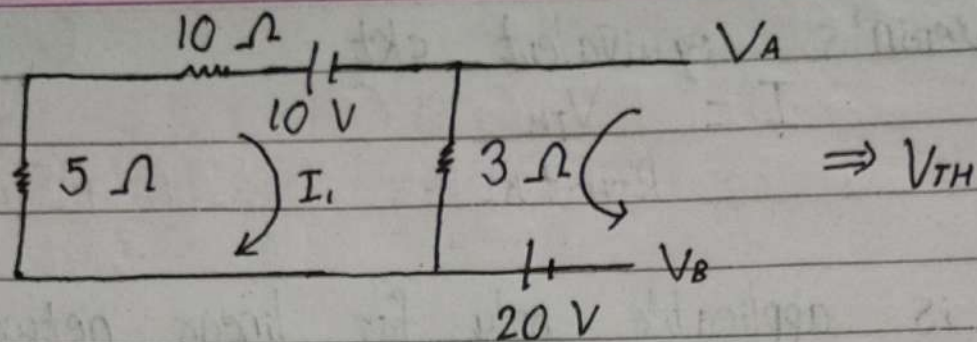
Solⁿ:

To find Thevenin's ^{equivalent} ~~load~~ resistance (R_{TH}),



$$R_{TH} = \frac{(5+10) \times 3}{(5+10)+3} = 2.5\ \Omega$$

To find Thevenin's equivalent voltage (V_{TH}),



$$V_{TH} - 3I_1 - 20 = 0$$

$$\text{or, } V_{TH} = 3I_1 + 20$$

For Mesh I_1 ,

$$-5I_1 - 10I_1 - 10 - 3I_1 = 0$$

$$\text{or, } -18I_1 - 10 = 0$$

$$\therefore I_1 = \frac{10}{18} = 0.5 \text{ A}$$

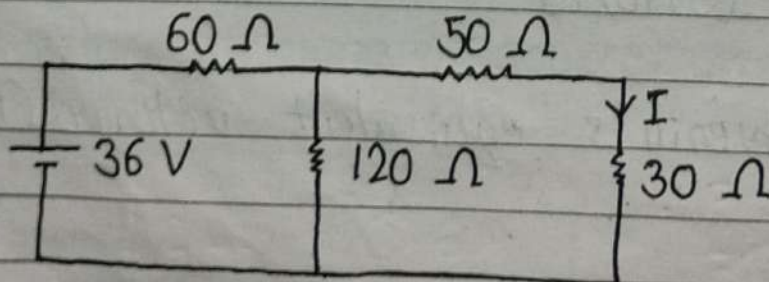
$$V_{TH} - 3I_1 - 20 = 0$$

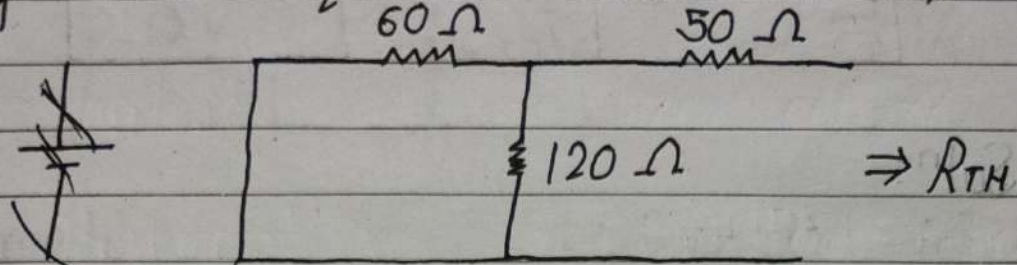
$$\text{or, } V_{TH} = 3I_1 + 20$$

$$\therefore V_{TH} = 3 \times 0.5 + 20 = 21.5 \text{ V}$$

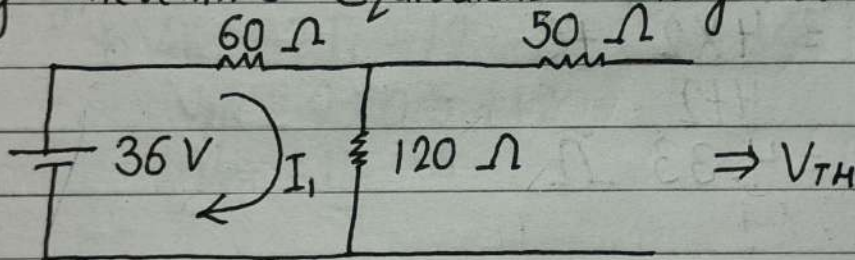
$$\text{Therefore, } I = \frac{21.5}{2.5 + 2} = 4.77 \text{ A} \#$$

(Q)



Solⁿ:Finding Thevenin's Equivalent Resistance (R_{TH}),

$$\begin{aligned}
 R_{TH} &= (60 \parallel 120) + 50 \\
 &= \frac{60 \times 120}{60 + 120} + 50 \\
 &= 90\ \Omega
 \end{aligned}$$

Finding Thevenin's Equivalent Voltage (V_{TH}),

For mesh,

$$-60I_1 - 120I_1 + 36 = 0$$

$$\text{or, } -180I_1 + 36 = 0$$

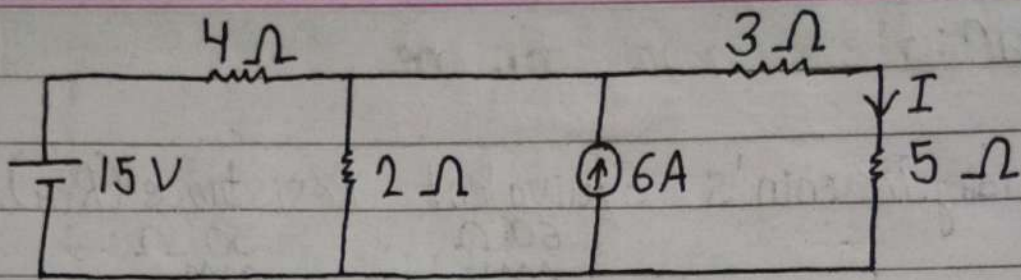
$$\therefore I_1 = \frac{36}{180} = 0.2\text{ A}$$

$$V_{TH} - 120I_1 = 0$$

$$\therefore V_{TH} = 120 \times 0.2 = 24\text{ V}$$

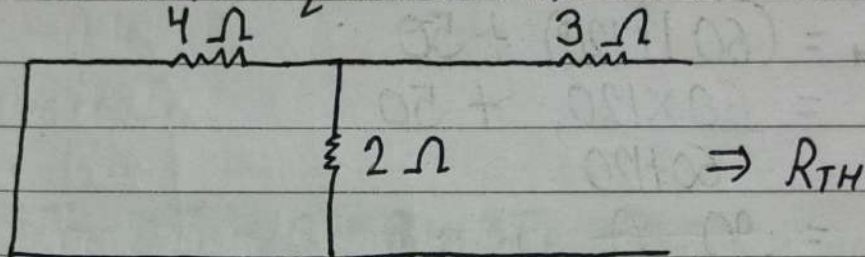
$$\text{Therefore, } I = \frac{24}{90 + 30} = 0.2\text{ A} \#$$

(Q)



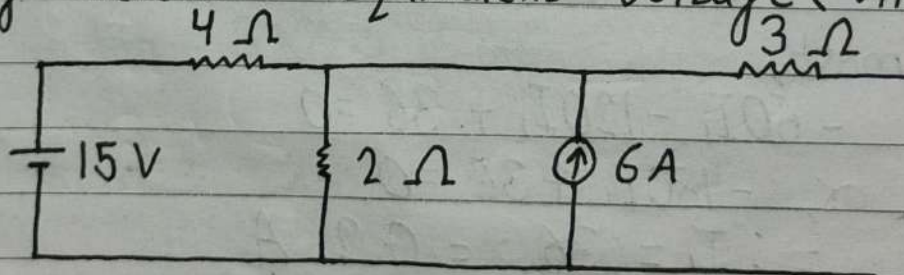
Solⁿ:

Finding Thevenin's Equivalent Resistance (R_{TH}),

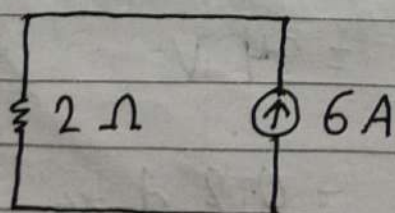


$$\begin{aligned} R_{TH} &= (4 \parallel 2) + 3 \\ &= \frac{4 \times 2}{4 + 2} + 3 \\ &= 4.33 \Omega \end{aligned}$$

Finding Thevenin's Equivalent Voltage (V_{TH}),

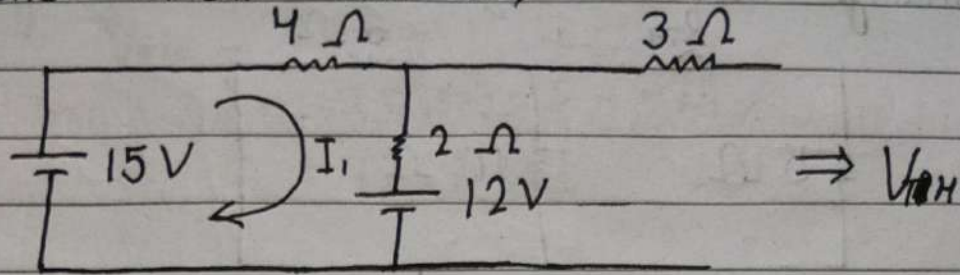


Converting current into voltage



$$\begin{aligned} V &= IR \\ &= 6 \times 2 \\ \therefore V &= 12 \text{ V} \end{aligned}$$

Then the circuit can be,



For mesh,

$$15 - 4I_1 - 2I_1 - 12 = 0$$

$$\text{or, } -6I_1 + 3 = 0$$

$$\text{or, } I_1 = \frac{3}{6}$$

$$\therefore I_1 = 0.5 \text{ A}$$

Similarly,

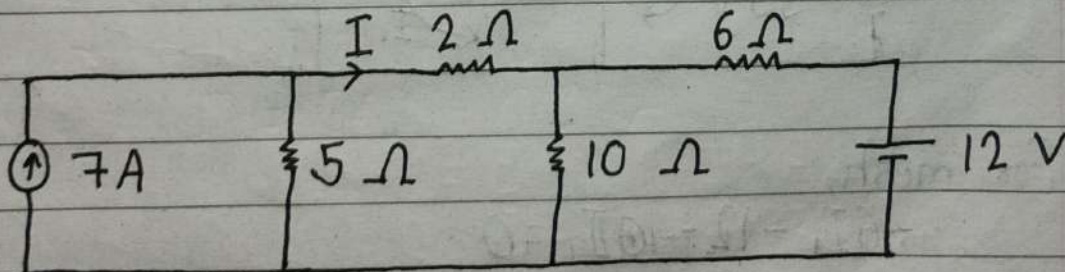
$$+V_{OC} - 2I_1 - 12 = 0$$

$$\text{or, } V_{OC} = 2 \times 0.5 + 12$$

$$\therefore V_{OC} = 13 \text{ V}$$

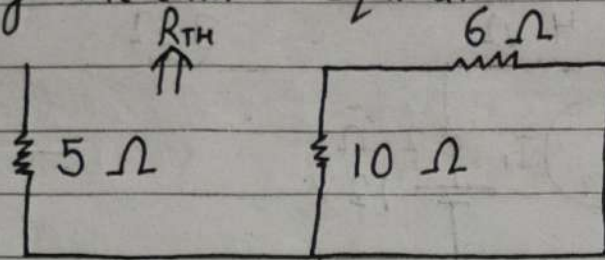
$$\text{Therefore, } I = \frac{13}{4.33 + 5} = 1.38 \text{ A} \neq$$

(9)



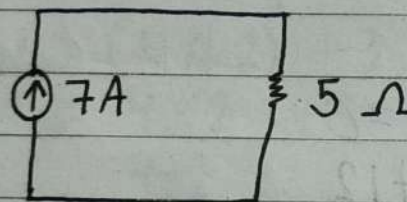
Soln:

Finding Thevenin's Equivalent Resistance (R_{TH}),



$$\begin{aligned} R_{TH} &= (6 \parallel 10) + 5 \\ &= \frac{6 \times 10}{6 + 10} + 5 \\ &= 8.75 \Omega \end{aligned}$$

Finding Thevenin's Equivalent Voltage (V_{TH}),

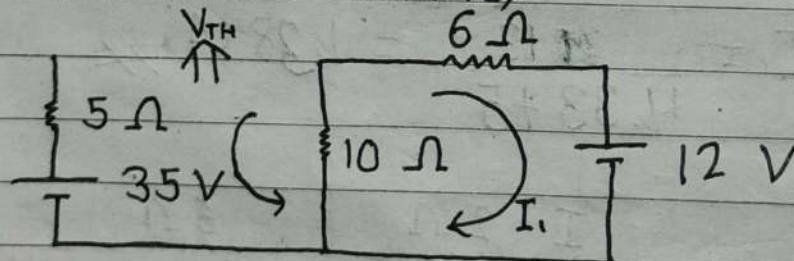


$$V = IR$$

$$= 7 \times 5$$

$$\therefore V = 35 \text{ V}$$

The circuit now became,



For mesh,

$$-6I_1 - 12 - 10I_1 = 0$$

$$\text{or, } -16I_1 - 12 = 0$$

$$\text{or, } I_1 = \frac{12}{16}$$

$$\therefore I_1 = 0.75 \text{ A}$$

Similarly,

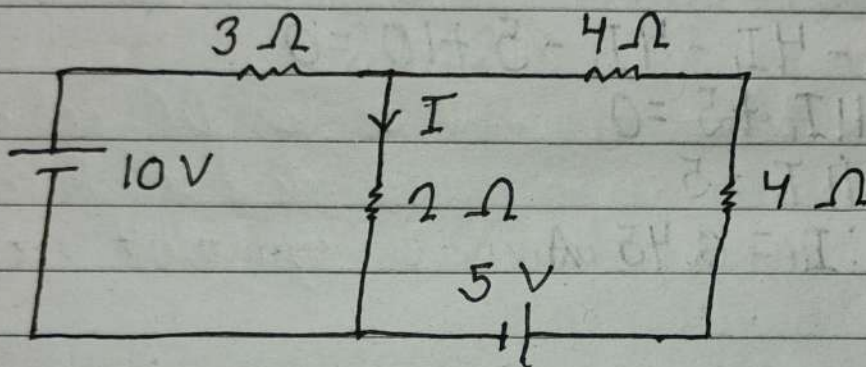
$$V_{TH} - 35 - 10I_1 = 0$$

$$\text{or, } V_{TH} = 35 + (10 \times 0.75)$$

$$\therefore V_{TH} = 42.5 \text{ V}$$

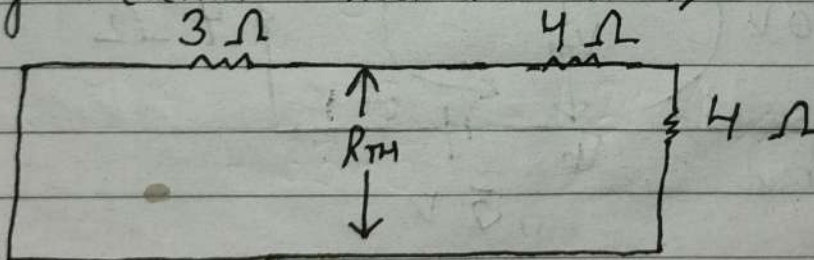
$$\text{Therefore, } I = \frac{42.5}{8.75 + 2} = 3.95 \text{ A} \#$$

9)



Solⁿ:

Finding Thevenin's Resistance (R_{TH}),

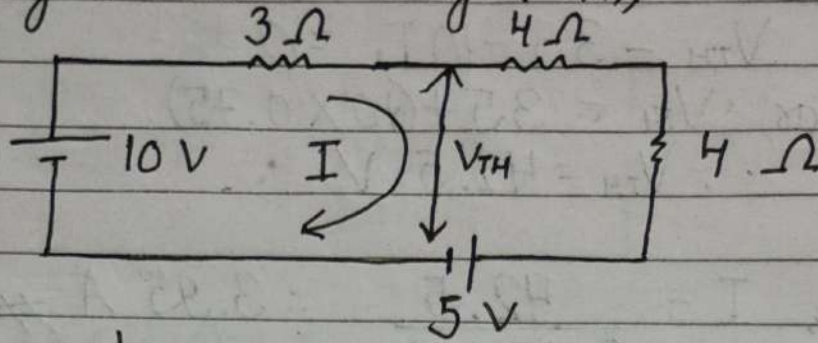


$$R_{TH} = (4+4) \parallel 3$$

$$= \frac{8 \times 3}{8+3}$$

$$= 2.1 \Omega$$

Finding Thevenin's Voltage (V_{TH}),



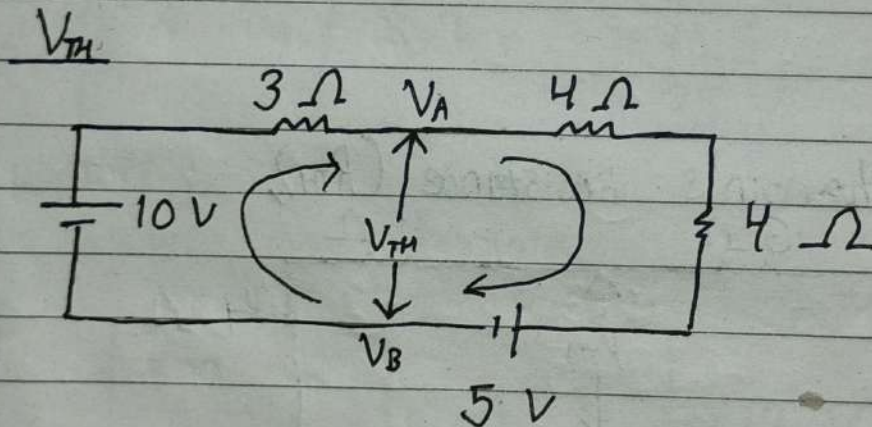
For mesh,

$$-3I_1 - 4I_1 - 4I_1 - 5 + 10 = 0$$

$$\text{or, } -11I_1 + 5 = 0$$

$$\text{or, } 11I_1 = 5$$

$$\therefore I_1 = 0.45 \text{ A}$$



(i) For second-half,

$$V_A - 4I_1 - 4I_1 - 5 = V_B$$

$$\text{or, } V_A - V_B - 8I_1 - 5 = 0$$

$$\text{or, } V_{TH} = 8 \times 0.45 + 5$$

$$\therefore V_{TH} = 8.6 \text{ V}$$

(ii) For first half,

$$V_B + 10 - 3I_1 = V_A$$

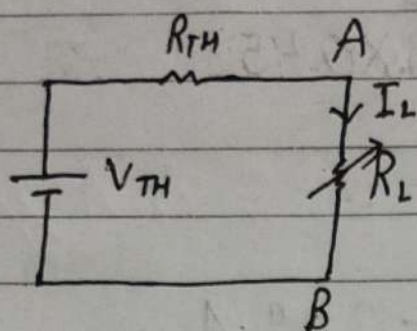
$$\text{or, } V_A - V_B = 10 - 3 \times 0.45$$

$$\therefore V_{TH} = 8.65 \text{ V}$$



$$\text{Therefore, } I = \frac{8.6}{2.1 + 2} = 2.09 \text{ A} \#$$

Imp # Maximum Power Transfer Theorem



In A resistive ^{load} ~~toode~~ connected to a D.C. network receive maximum power when load resistance R_L is equal to the Thevenin's equivalent resistance R_{TH} for a D.C. network.

$$\text{i.e. } R_{TH} = R_L$$

Applications:

- It is used in communication system, electronic circuit, loud speakers, etc.

Let, current through the R_L be I_L .

$$I_L = \frac{V_{TH}}{R_{TH} + R_L} \quad \text{--- (1)}$$

Power developed across load

$$P_L = I_L^2 R_L \quad \text{--- (2)}$$

$$P_L = \frac{V_{TH}^2}{(R_{TH} + R_L)^2} \times R_L \quad \text{--- (3)}$$

For maximum power ^{across} resistive load P_{max} is,

$$P_{max} = \frac{d(P_L)}{dR_L} = 0$$

$$\frac{d}{dR_L} \left[\frac{V_{TH}^2}{(R_{TH} + R_L)^2} \times R_L \right] = 0$$

$$\text{or, } \frac{(R_{TH} + R_L)^2 \cdot V_{TH}^2 - V_{TH}^2 \cdot R_L \cdot 2(R_{TH} + R_L)}{(R_{TH} + R_L)^4} = 0$$

$$\text{or, } V_{TH}^2 [(R_{TH} + R_L)^2 - 2R_L(R_{TH} + R_L)] = 0$$

$$\text{or, } (R_{TH} + R_L)^2 = 2R_L(R_{TH} + R_L)$$

$$\text{or, } R_{TH} + R_L = 2R_L$$

$$\text{or, } R_{TH} = 2R_L - R_L$$

$$\therefore \boxed{R_{TH} = R_L}$$

So,

$$P_{max} = \frac{V_{TH}^2}{(R_{TH} + R_{TH})^2}$$

$$\therefore \boxed{P_{max} = \frac{V_{TH}^2}{4R_{TH}}}$$

Efficiency of maximum power theorem

Efficiency is the ratio of maximum power across the load to the total power of the circuit.

$$\eta\% = \frac{P_{\text{Load}}}{P_{\text{Total}}}$$

Power across R_{TH} ,

$$P_H = \frac{V_{TH}^2}{4R_L}$$

Power across R_L ,

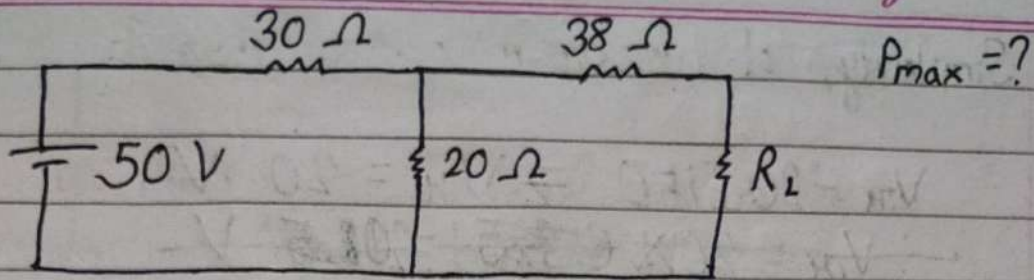
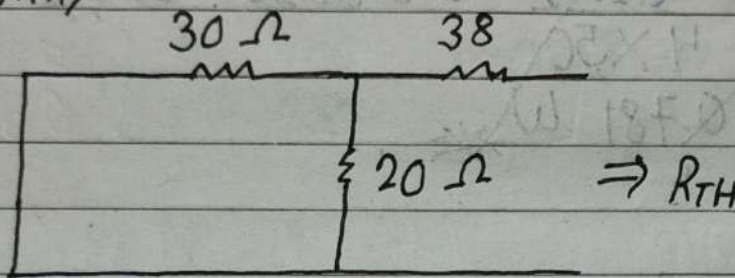
$$P_L = \frac{V_{TH}^2}{4R_L}$$

$$\begin{aligned} P_{\text{Total}} = P_{TH} + P_L &= \frac{V_{TH}^2}{4R_L} + \frac{V_{TH}^2}{4R_L} \\ &= \frac{V_{TH}^2}{2R_L} \end{aligned}$$

$$\eta\% = \frac{V_{TH}^2 / 4R_L}{V_{TH}^2 / 2R_L}$$

$$\therefore \eta = 50\%$$

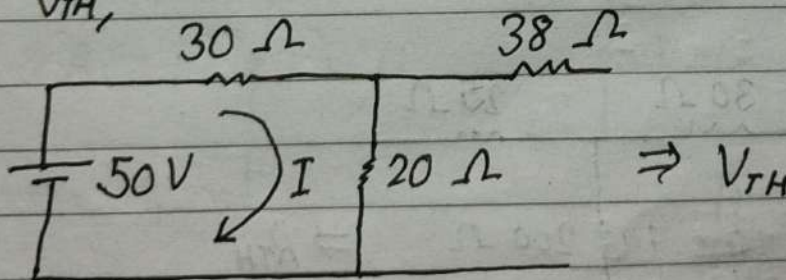
Q)

Solⁿ:For R_{TH} ,

$$R_{TH} = \left(\frac{30 \times 20}{30 + 20} \right) + 38$$

$$= 50 \, \Omega$$

i.e. $R_L = R_{TH} = 50 \, \Omega$

For V_{TH} ,

For mesh,

$$50 - 30I - 20I = 0$$

$$\therefore 50 - 50I = 0$$

$$\therefore I = \frac{50}{50} = 1 \, A$$

Similarly,

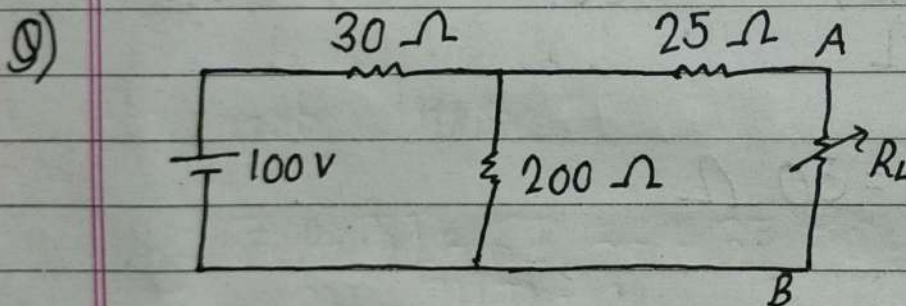
$$V_{TH} - 20I = 0 \Rightarrow V_{TH} = 20 \text{ V}$$

$$\therefore V_{TH} = 20 \times \cancel{0.0625} = \cancel{201.25} \text{ V}$$

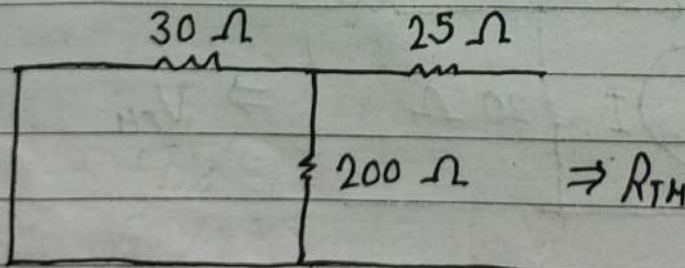
Therefore, the maximum power P_{max} is,

$$P_{max} = \frac{(12.5)^2}{4 \times 50} = \cancel{0.0625 \text{ W}} \\ = \cancel{0.781 \text{ W}}$$

$$P_{max} = \frac{20^2}{4 \times 50} = 2 \text{ W}$$

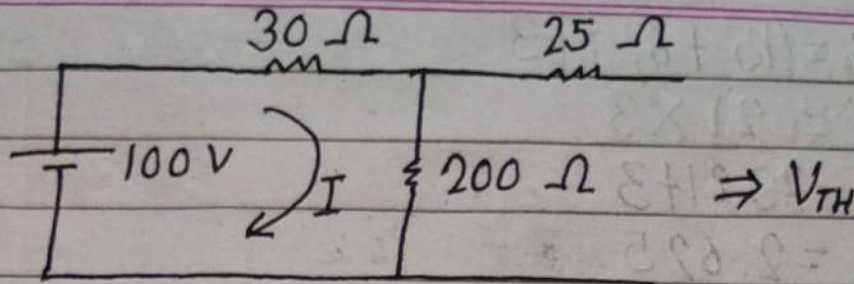


Solⁿ:



$$R_{TH} = (30 \parallel 200) + 25 \\ = 51.08 \Omega$$

$$\text{i.e. } R_{TH} = R_L = 51.08 \Omega$$



For mesh,

$$100 - 30I - 200I = 0$$

$$\text{or, } 100 - 230I = 0$$

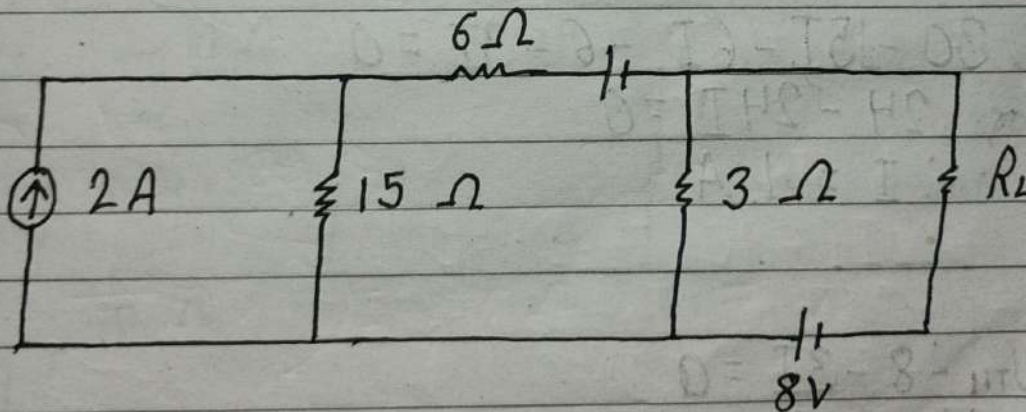
$$\therefore I = \frac{100}{230} = 0.43 \text{ A}$$

$$V_{TH} - 200I = 0$$

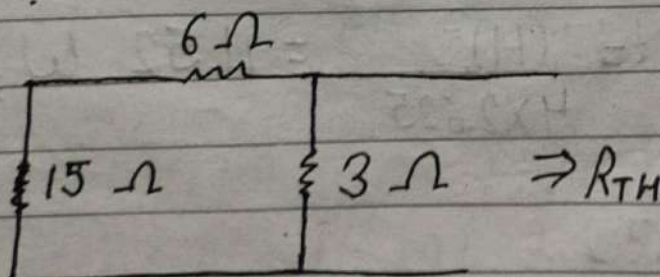
$$\therefore V_{TH} = 200 \times 0.43 = 86.95 \text{ V}$$

$$\therefore P_{\max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{(86.95)^2}{4 \times 51.08} = 37 \text{ W} \#$$

Q)

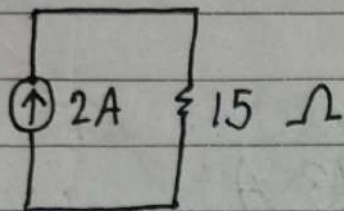


Solⁿ:

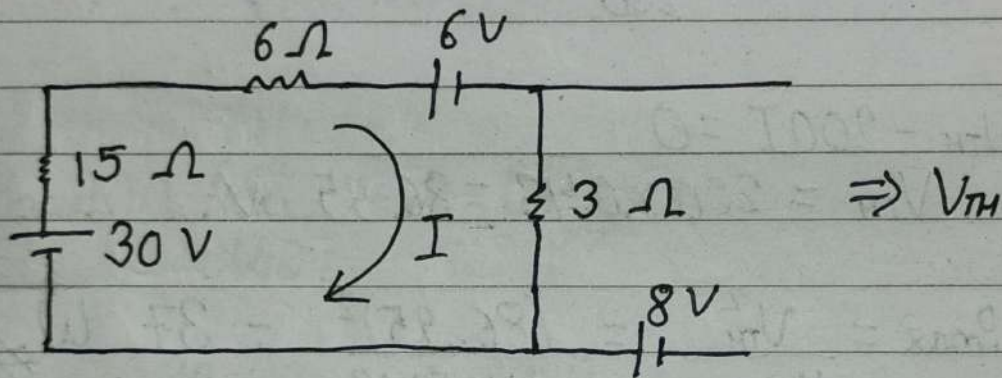


$$R_{TH} = 6 \parallel 15$$

$$\begin{aligned}
 R_{TH} &= (15 + 6) \parallel 3 \\
 &= \frac{21 \times 3}{21 + 3} \\
 &= 2.625 \\
 &= R_L
 \end{aligned}$$



$$V = 2 \times 15 = 30 \text{ V}$$



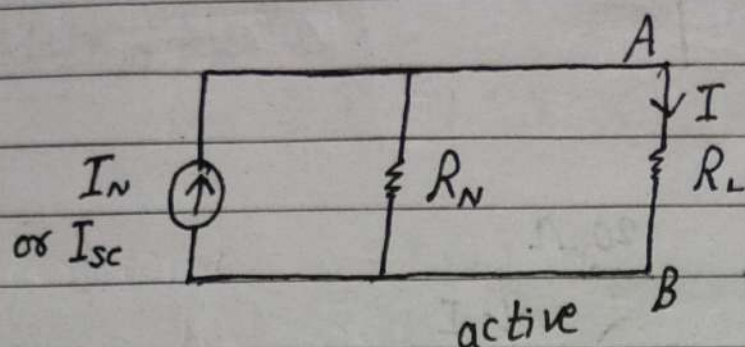
For mesh,

$$\begin{aligned}
 30 - 15I - 6I - 6 - 3I &= 0 \\
 \therefore 24 - 24I &= 0 \\
 \therefore I &= 1 \text{ A}
 \end{aligned}$$

$$\begin{aligned}
 V_{TH} - 8 - 3I &= 0 \\
 \therefore V_{TH} &= 8 + 3 \times 1 = 11 \text{ V}
 \end{aligned}$$

$$\therefore P_{\max} = \frac{V_{TH}^2}{4R_L} = \frac{(11)^2}{4 \times 2.625} = 11.52 \text{ W} \#$$

Norton's Theorem

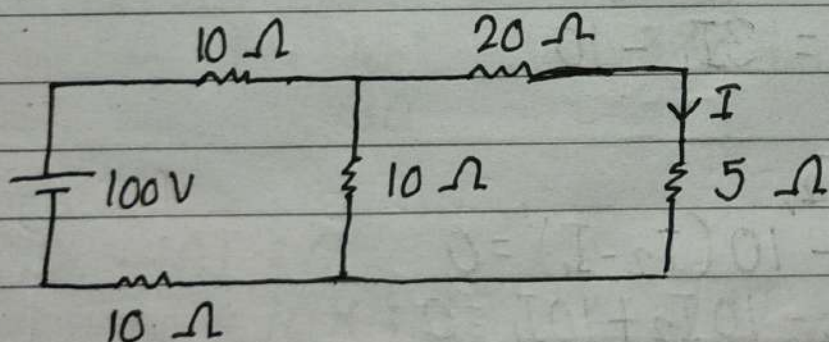


A linear bilateral D.C. network containing Norton's current (I_N) or short-circuit current (I_{sc}) in parallel with Norton's resistance (R_N), where I_N is the I_{sc} across the load resistance (R_L).

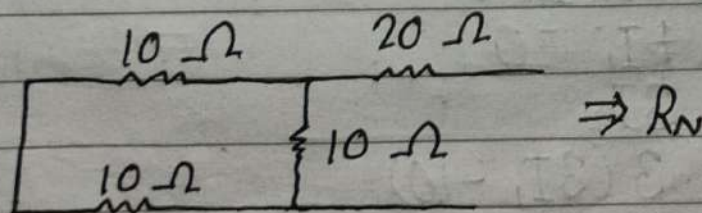
Where energy sources are replaced to their internal resistance.

$$I = (I_N \text{ or } I_{sc}) \times \frac{R_N}{R_N + R_L}$$

Q)



Solⁿ:



$$R_N = \left[\frac{(10 \parallel 10)}{10+10} \right] + 20 \quad R_N = [(10+10) \parallel 10] + 20$$

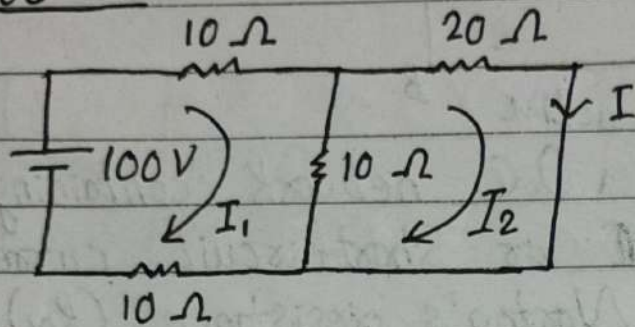
$$= \left[\frac{10 \times 10}{10+10} \right] + 20 \quad = \frac{20 \times 10}{20+10} + 20$$

Page: _____

$$= \frac{5 \times 10}{5 + 10} + 20 \quad \therefore R_N = 26.66 \, \Omega$$

$$= 23.33$$

Method - 1st



For mesh I_1 ,

$$100 - 10I_1 - 10I_1 - 10(I_1 - I_2) = 0$$

$$\text{or, } 100 - 20I_1 - 10I_1 + 10I_2 = 0$$

$$\text{or, } 100 - 30I_1 + 10I_2 = 0$$

$$\text{or, } 10I_2 = 30I_1 - 100$$

$$\text{or, } I_2 = \frac{30I_1 - 100}{10}$$

$$\text{or, } I_2 = 3I_1 - 10$$

For mesh I_2 ,

$$-20I_2 - 10(I_2 - I_1) = 0$$

$$\text{or, } -20I_2 - 10I_2 + 10I_1 = 0$$

$$\text{or, } -30I_2 + 10I_1 = 0$$

$$\text{or, } -3I_2 + I_1 = 0$$

$$\text{or, } I_1 = 3I_2$$

$$\text{or, } I_1 = 3(3I_1 - 10)$$

$$\text{or, } I_1 = 9I_1 - 30$$

$$\text{or, } 30 = 8I_1$$

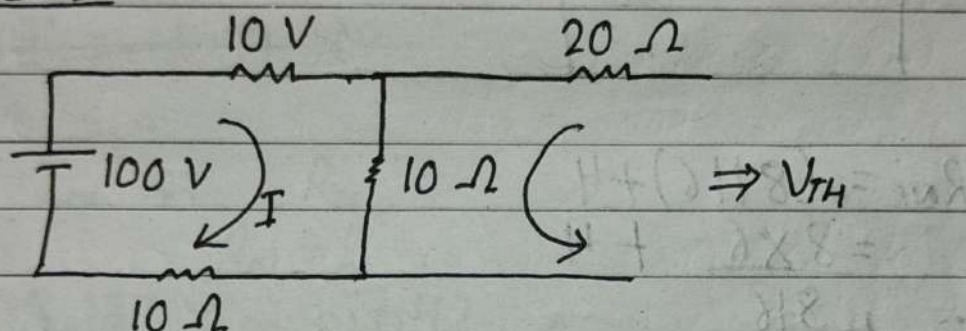
$$\therefore I_1 = 3.75 \, \text{A}$$

$$\therefore I_2 = 3 \times 3.75 - 10 = 1.25 \text{ A i.e. } I_N = 1.25 \text{ A}$$

~~Method - 2nd~~

$$\therefore I = 1.25 \times \frac{26.66}{26.66 + 5} = 1.05 \text{ A} \#$$

Method - 2nd



For mesh I,

$$100 - 10I - 10I - 10I = 0$$

$$\text{or, } 100 - 30I = 0$$

$$\therefore I = \frac{100}{30} = 3.33 \text{ A}$$

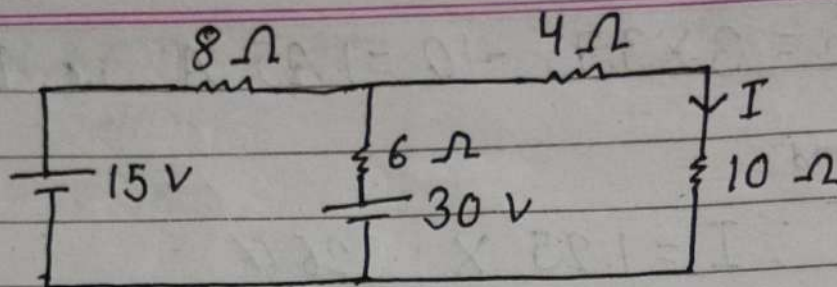
$$V_{TH} - 10I = 0$$

$$\therefore V_{TH} = 10 \times 3.33 = 33.3 \text{ V}$$

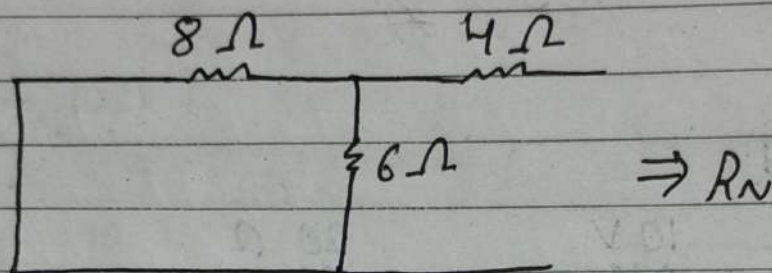
$$\therefore I_N = \frac{V_{TH}}{R_{TH}} = \frac{33.3}{26.66} = 1.249 \text{ A}$$

$$\text{Therefore, } I = 1.249 \times \frac{26.66}{26.66 + 5} = 1.05 \text{ A} \#$$

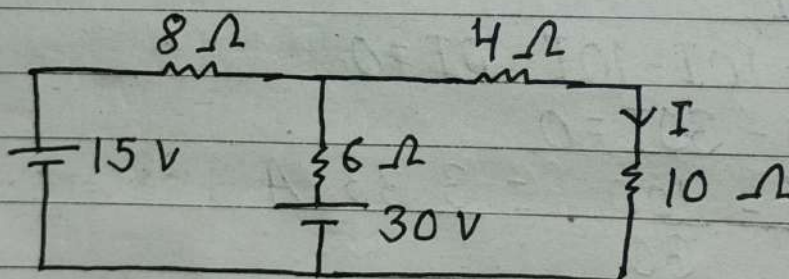
9)



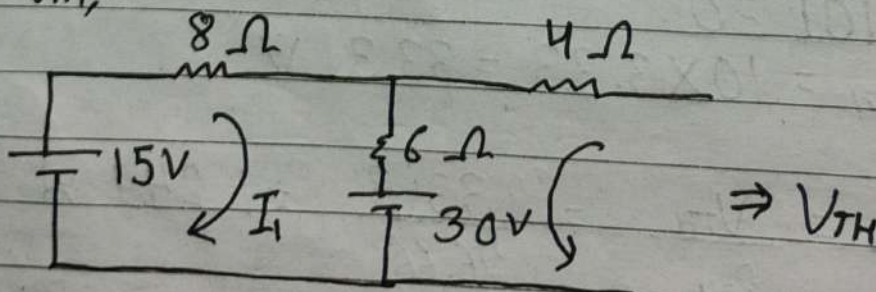
Solⁿ:



$$\begin{aligned}
 R_N &= (8 \parallel 6) + 4 \\
 &= \frac{8 \times 6}{8 + 6} + 4 \\
 &= 7.42 \, \Omega
 \end{aligned}$$



For V_{TH} ,



For mesh,

$$\begin{aligned}
 15 - 8I_1 - 6I_1 - 30 &= 0 \\
 \text{or, } -15 - 14I_1 &= 0
 \end{aligned}$$

$$\therefore I_1 = \frac{-15}{14} = -1.07 \text{ A}$$

$$V_{TH} - 6I_1 - 30 = 0$$

$$\therefore V_{TH} = 6 \times (-1.07) + 30 = 23.58 \text{ V}$$

Therefore,

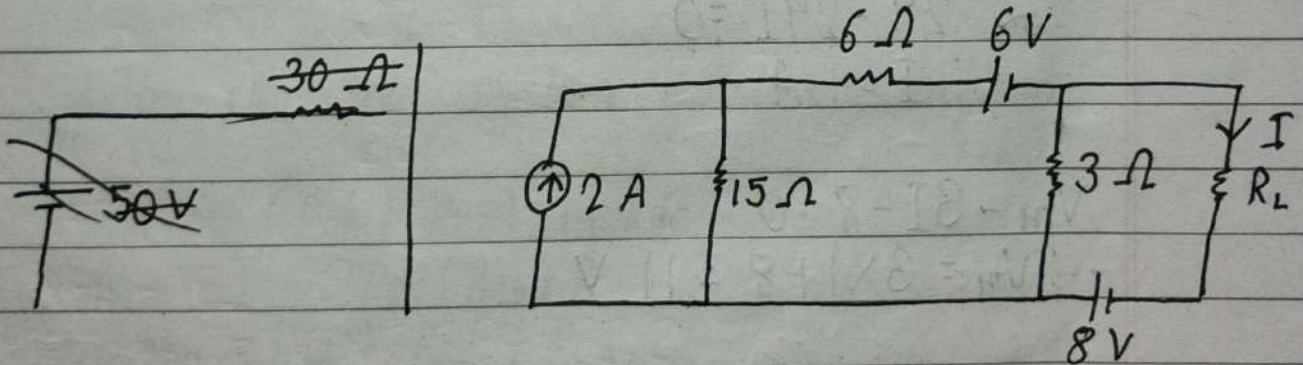
~~$$I = 23.58$$~~

$$I_N = \frac{V_{TH}}{R_{TH}} = \frac{23.58}{7.42} = 3.177 \text{ A}$$

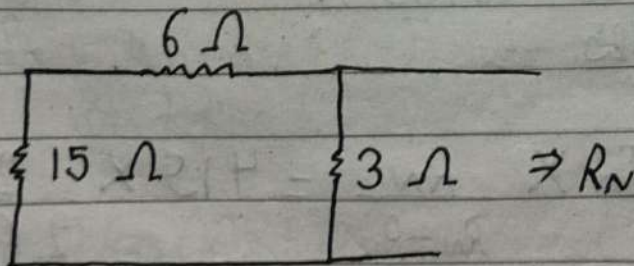
~~$$I = 1.07 \text{ A}$$~~

$$\begin{aligned} I &= I_N \times \frac{R_N}{R_N + R_L} \\ &= 3.177 \times \frac{7.42}{7.42 + 10} \\ &= 1.353 \text{ A} \end{aligned}$$

9)



Solⁿ:

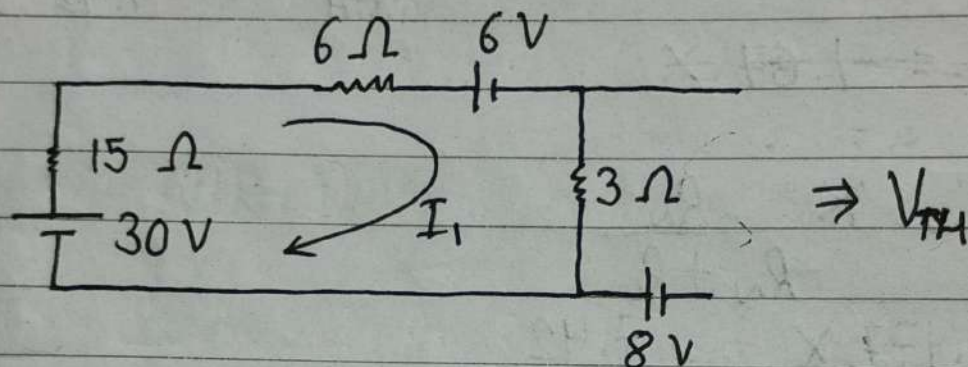
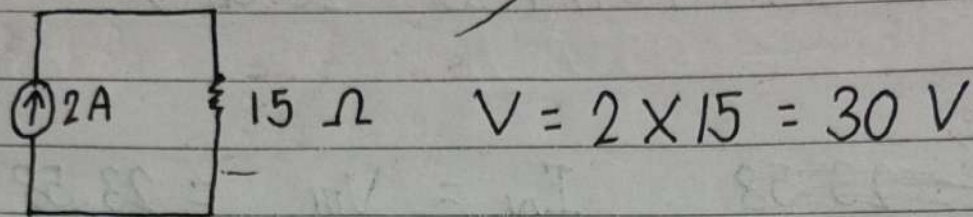


$$R_N = (15 + 6) \parallel 3$$

$$R_N = \frac{21 \times 3}{21+3}$$

$$= 2.625 \, \Omega$$

$$= R_L$$



For mesh I_1 ,

$$30 - 15I_1 - 6I_1 - 6 - 3I_1 = 0$$

$$\text{or, } 24 - 24I_1 = 0$$

$$\therefore I_1 = 1 \text{ A}$$

$$V_{TH} - 3I_1 - 8 = 0$$

$$\therefore V_{TH} = 3 \times 1 + 8 = 11 \text{ V}$$

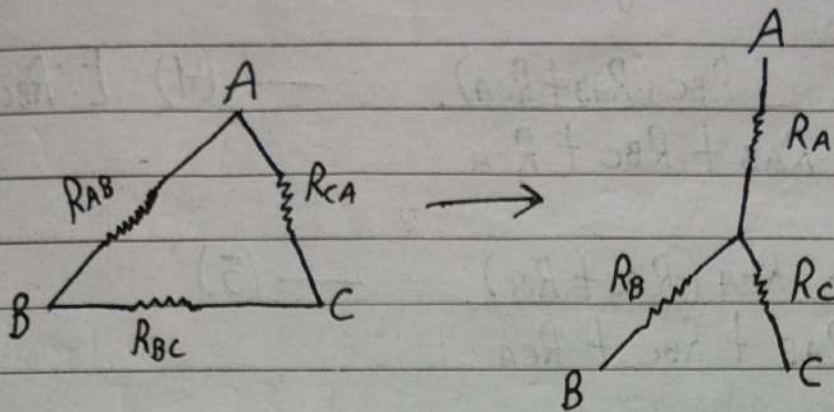
$$I_N = \frac{V_{TH}}{R_N} = \frac{11}{2.625} = 4.19 \text{ A}$$

Therefore, $I = I_N \times \frac{R_N}{R_N + R_L} = 4.19 \times \frac{2.625}{2.625 + 2}$

$$\therefore I = 2.09 \text{ A} \quad \#$$

Imp

Delta-to-Star Transformation



$$R_A = \frac{R_{AB} R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_B = \frac{R_{AB} R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_C = \frac{R_{BC} R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

In delta connected system, resistance between point B & C,

$$\begin{aligned} R_{BC} &= (R_{AB} + R_{CA}) \parallel R_{BC} \\ &= \frac{R_{BC} (R_{AB} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (1)} \end{aligned}$$

Similarly,

$$R_{CA} = \frac{R_{CA} (R_{AB} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (2)}$$

$$R_{AB} = \frac{R_{AB} (R_{CA} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (3)}$$

In star connected system,

$$R_B + R_C = \frac{R_{BC} (R_{AB} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (4)} \quad [\because R_{BC} = R_B + R_C]$$

$$R_C + R_A = \frac{R_{CA} (R_{AB} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (5)}$$

$$R_A + R_B = \frac{R_{AB} (R_{CA} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (6)}$$

At eqⁿ. (4), (5) & (6), adding,

$$2(R_A + R_B + R_C) = \frac{2(R_{AB} R_{BC} + R_{BC} R_{CA} + R_{CA} R_{AB})}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_A + R_B + R_C = \frac{R_{AB} R_{BC} + R_{BC} R_{CA} + R_{CA} R_{AB}}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (7)}$$

Eqⁿ. (7) - (4),

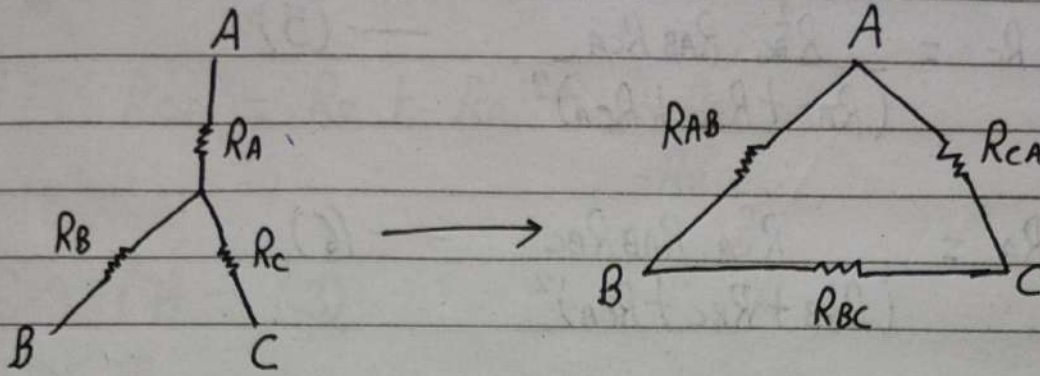
$$R_A = \frac{R_{AB} R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

Similarly, (7) - (5),

$$R_B = \frac{R_{AB} R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_C = \frac{R_{AC} R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

Imp # Star-to-Delta Transformation



$$R_{AB} = R_A + R_B + \frac{R_A R_B}{R_C}$$

$$R_{BC} = R_B + R_C + \frac{R_B R_C}{R_A}$$

$$R_{CA} = R_C + R_A + \frac{R_C R_A}{R_B}$$

$$R_A = \frac{R_{AB} R_{CA}}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (1)}$$

$$R_B = \frac{R_{AB} R_{BC}}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (2)}$$

$$R_C = \frac{R_{BC} R_{CA}}{R_{AB} + R_{BC} + R_{CA}} \quad \text{--- (3)}$$

$$R_A R_B = \frac{R_{AB}^2 R_{BC} R_{CA}}{(R_{AB} + R_{BC} + R_{CA})^2} \quad \text{--- (4)}$$

$$R_B \cdot R_C = \frac{R_{BC}^2 \cdot R_{AB} R_{CA}}{(R_{AB} + R_{BC} + R_{CA})^2} \quad \text{--- (5)}$$

$$R_C \cdot R_A = \frac{R_{CA}^2 \cdot R_{AB} R_{BC}}{(R_{AB} + R_{BC} + R_{CA})^2} \quad \text{--- (6)}$$

Adding eq^{ns}. (4), (5) & (6),

$$\begin{aligned} R_A R_B + R_B R_C + R_C R_A &= \frac{(R_{AB}^2 R_{BC} R_{CA}) + (R_{BC}^2 R_{AB} R_{CA}) + (R_{CA}^2 R_{AB} R_{BC})}{(R_{AB} + R_{BC} + R_{CA})^2} \\ &= \frac{R_{AB} R_{BC} R_{CA} (R_{AB} + R_{BC} + R_{CA})}{(R_{AB} + R_{BC} + R_{CA})^2} \end{aligned}$$

~~Multiplying eqⁿ. (4), (5) & (6),~~

~~RAR~~

$$R_A R_B + R_B R_C + R_C R_A = \frac{R_{AB} R_{BC} R_{CA}}{(R_{AB} + R_{BC} + R_{CA})} \quad \text{--- (7)}$$

Eqⁿ. (7) $\div$ (1),

$$\frac{R_A R_B}{R_A} + \frac{R_B R_C}{R_A} + \frac{R_C R_A}{R_A} = \frac{R_{AB} R_{BC} R_{CA} / (R_{AB} + R_{BC} + R_{CA})}{R_{AB} R_{CA} / (R_{AB} + R_{BC} + R_{CA})}$$

$$R_B + \frac{R_B R_C}{R_A} + R_C = R_{BC}$$

which is required.

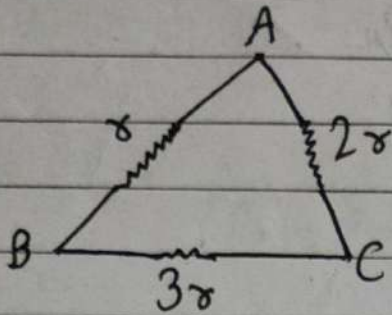
Exⁿ. (7) $\div$ (2),

$$R_{CA} = R_C + R_A + \frac{R_C R_A}{R_B}$$

Exⁿ. (7) $\div$ (3),

$$R_{AB} = R_A + R_B + \frac{R_A R_B}{R_C}$$

Q) Find R_A, R_B & R_C ?

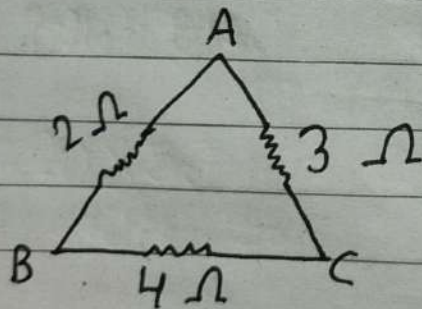


$$R_A = \frac{x \times 2x}{x + 2x + 3x} = \frac{2x^2}{6x} = \frac{x}{3}$$

$$R_B = \frac{x \times 3x}{6x} = \frac{x}{2}$$

$$R_C = \frac{3x \times 2x}{6x} = x$$

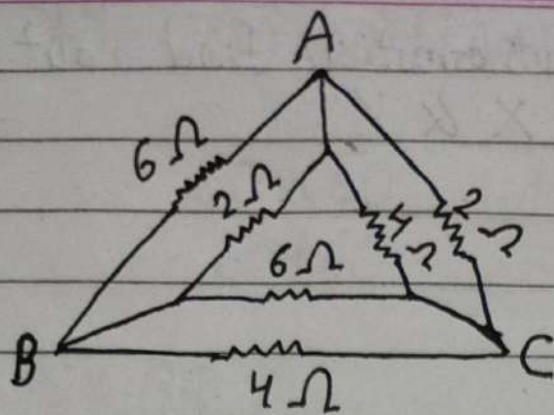
Q) Find resistance between B & C?



$$\begin{aligned} R_{BC} &= (3+2) \parallel 4 \\ &= \frac{5 \times 4}{5+4} \\ &= \frac{20}{9} \end{aligned}$$

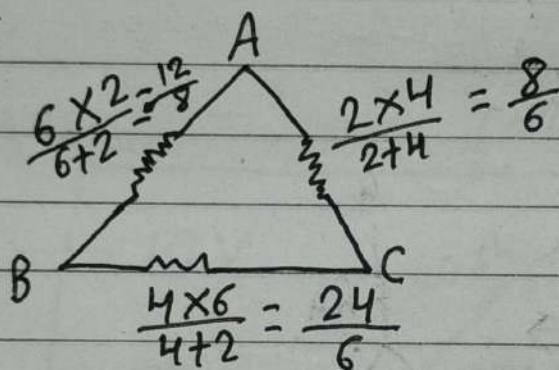
$$= 2.22 \Omega \#$$

Q)



Find resistance between point B & C?

Solⁿ:

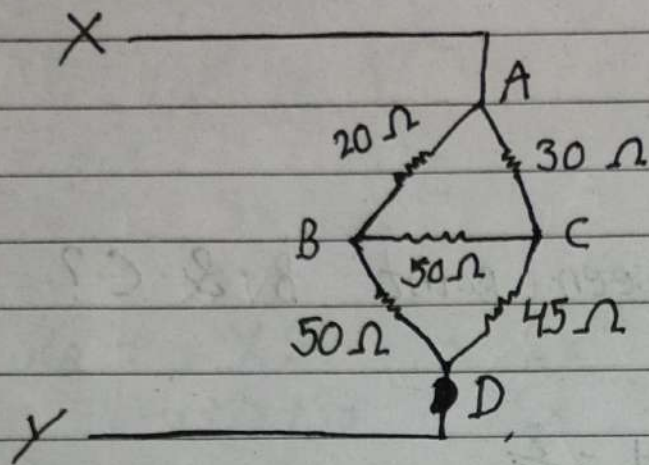


$$R_{BC} = \frac{24/6 \left(\frac{12}{8} + \frac{8}{6} \right)}{24/6 + 12/8 + 8/6}$$

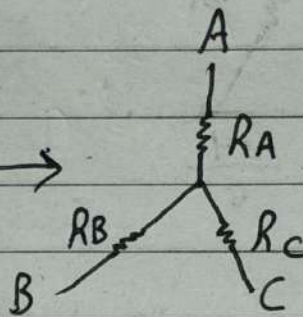
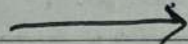
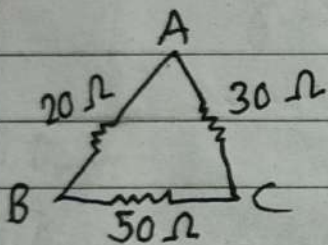
$$= \frac{11.33}{6.83}$$

$$= 1.65 \Omega \#$$

9) Using star-to-delta transformation, find out the resistance between point X & Y.



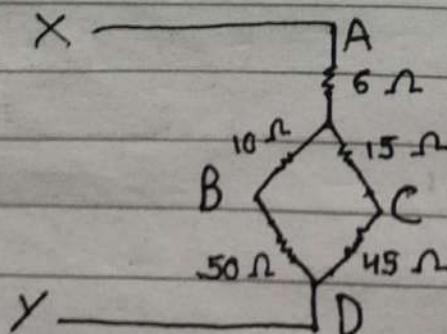
Solⁿ:



$$R_A = \frac{20 \times 30}{20 + 30 + 50} = \frac{600}{100} = 6 \, \Omega$$

$$R_B = \frac{20 \times 50}{100} = 10 \, \Omega$$

$$R_C = \frac{50 \times 30}{100} = 15 \, \Omega$$



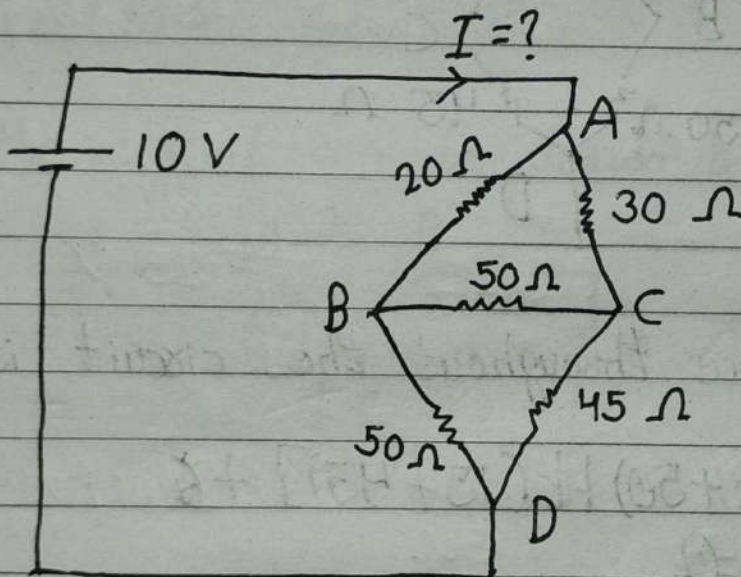
$$R_{xy} = [(10+50) \parallel (15+45)] + 6$$

$$= \frac{60 \times 60}{60+60} + 6$$

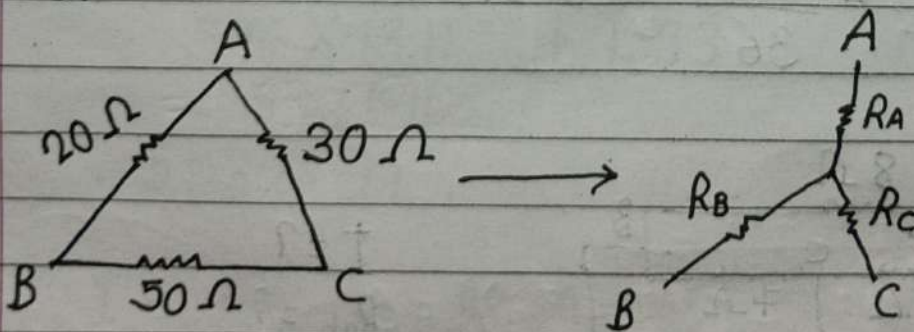
$$= 36 \Omega$$

Therefore, the resistance between point X & Y is 36Ω .

Q)



Solⁿ:

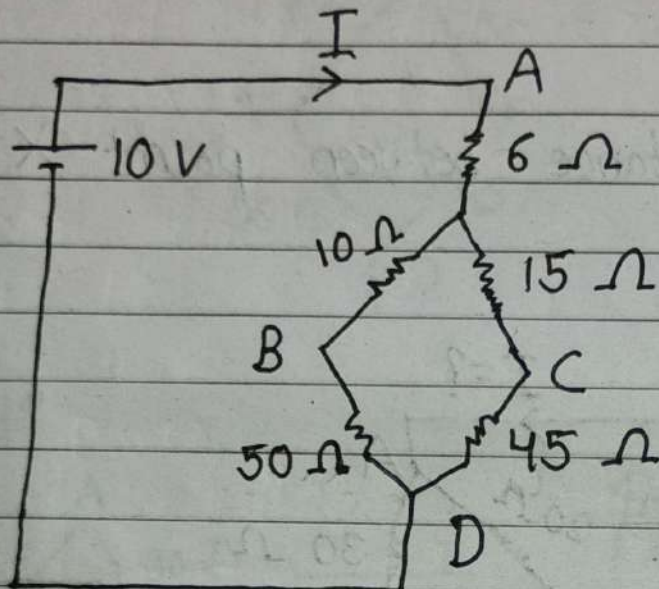


$$R_A = \frac{20 \times 30}{20+30+50} = 6 \Omega$$

$$R_B = 10 \, \Omega$$

$$R_C = 15 \, \Omega$$

Now, the circuit can be drawn as;



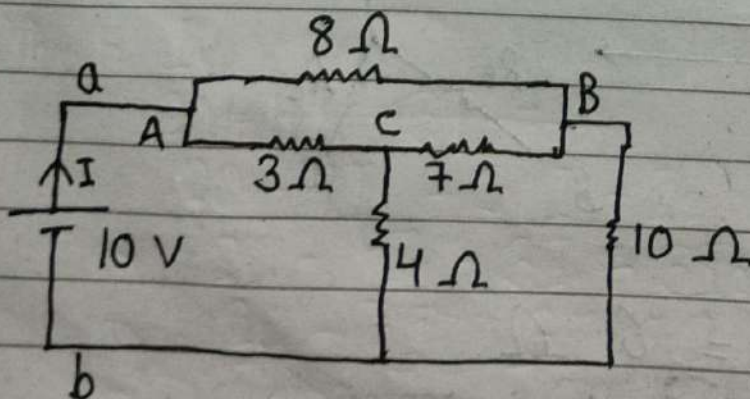
The resistance throughout the circuit is given as;

$$[(10+50) \parallel (15+45)] + 6$$

$$= 36 \, \Omega$$

Therefore, $I = \frac{10}{36} = 0.277 \, A$

9)

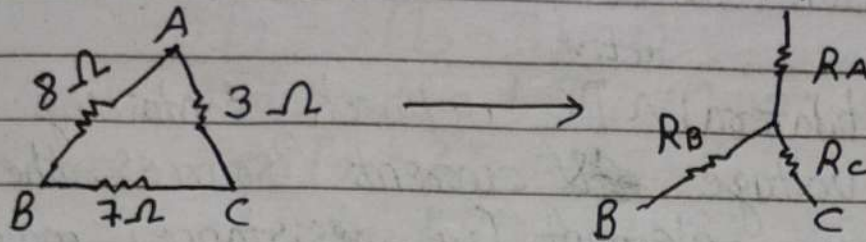


$$I = ?$$

$$R_{ab} = ?$$

$$P = ?$$

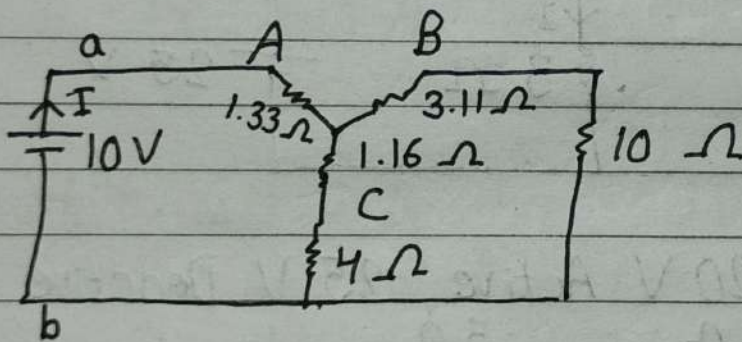
Soln:



$$R_A = \frac{8 \times 3}{8 + 7 + 3} = 1.33 \, \Omega \quad R_C = \frac{7 \times 3}{8 + 7 + 3} = 1.16 \, \Omega$$

$$R_B = \frac{8 \times 7}{8 + 7 + 3} = 3.11 \, \Omega$$

Now, the circuit can be drawn as:



$$\begin{aligned} R_{ab} &= [(1.16 + 4) \parallel (3.11 + 10)] + 1.33 \\ &= \frac{5.16 \times 13.11}{5.16 + 13.11} + 1.33 \\ &= 5.032 \, \Omega \end{aligned}$$

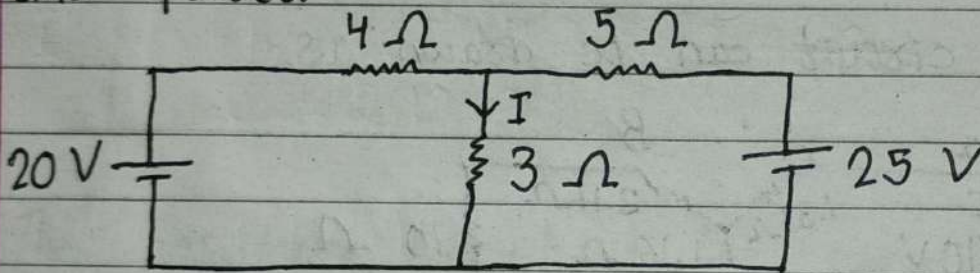
$$\therefore I = \frac{10}{5.032} = 1.98 \, A$$

$$P = \frac{10^2}{5.032} = 19.87 \, W$$

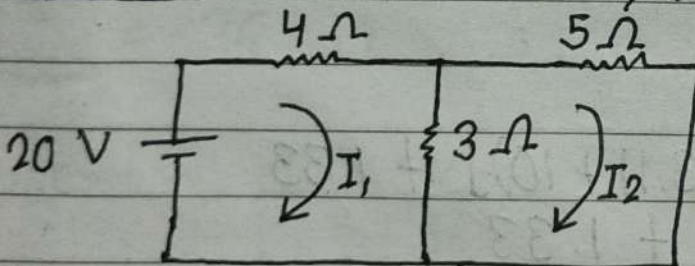
Super-position Theorem

A linear bilateral ^{active} DC network containing two or more voltage ~~and~~ current sources, the current through any element (i.e. resistance) may be determined by adding together algebraic sum, when current produced each source separately, when energy sources are replaced to their internal resistances.

It is used for measurement of voltage, current and power.



Case - I^{st} : 20 V Active, 25 V Deactive



Mesh I_1 ,

$$\cancel{20} - 4I_1 - 3I_1$$

$$20 - 4I_1 - 3(I_1 - I_2) = 0$$

$$\text{or, } 20 - 4I_1 - 3I_1 + 3I_2 = 0$$

$$\text{or, } 20 - 7I_1 + 3I_2 = 0$$

$$\text{or, } I_2 = \frac{7I_1 - 20}{3} \quad \text{--- (1)}$$

3

Mesh I_2 ,

$$-5I_2 - 3(I_2 - I_1) = 0$$

$$\text{or, } -5I_2 - 3I_2 + 3I_1 = 0$$

$$\text{or, } -8I_2 + 3I_1 = 0$$

$$\text{or, } -8\left(\frac{7I_1 - 20}{3}\right) + 3I_1 = 0$$

$$\text{or, } -56I_1 + 160 + 9I_1 = 0$$

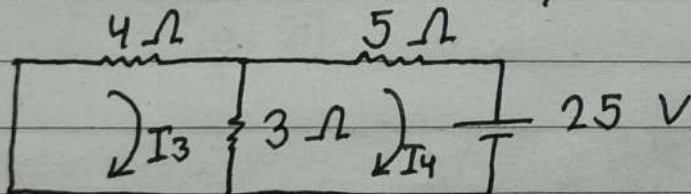
$$\text{or, } +47I_1 = +160$$

$$\therefore I_1 = 3.4 \text{ A}$$

$$I_2 = \frac{7 \times 3.4 - 20}{3} = 1.26 \text{ A}$$

$$I' = I_1 - I_2 = 2.14 \text{ A}$$

Case - IInd: 20 V Deactive, 25V Activate



Mesh I_3 ,

$$-4I_3 - 3(I_3 - I_4) = 0$$

$$\text{or, } -4I_3 - 3I_3 + 3I_4 = 0$$

$$\text{or, } -7I_3 + 3I_4 = 0$$

$$\text{or, } I_4 = \frac{7I_3}{3}$$

Mesh I_4 ,

$$-5I_4 - 25 - 3(I_4 - I_3) = 0$$

$$\text{or, } -5I_4 - 25 - 3I_4 + 3I_3 = 0$$

$$\text{or, } -8I_4 - 25 + 3I_3 = 0$$

$$\text{or, } -8\left(\frac{7I_3}{3}\right) - 25 + 3I_3 = 0$$

$$\text{or, } -56I_3 - 75 + 9I_3 = 0$$

$$\text{or, } -47I_3 - 75 = 0$$

$$\text{or, } -47I_3 = 75$$

$$\therefore I_3 = -1.59 \text{ A}$$

$$I_4 = \frac{-7 \times 1.59}{3} = -3.71 \text{ A}$$

$$I'' = I_3 - I_4 = -1.59 + 3.71 = 2.12 \text{ A}$$

$$I = 2.14 + 2.12 = 4.26 \text{ A} \quad \underline{\underline{\text{Ans}}}$$

Millman's Theorem

This theorem is helpful in finding a single equivalent voltage source for a number of voltage sources connected in parallel. It states that, 'n' number of parallel voltage sources $V_1, V_2, V_3, \dots, V_n$ with internal resistances $R_1, R_2, R_3, \dots, R_n$ respectively as shown in fig. (a) can be replaced by a single voltage source V in series with equivalent resistance as shown in fig. (b).

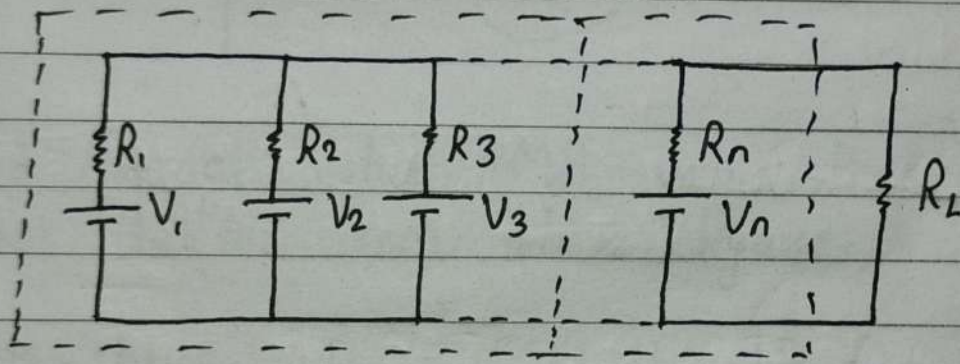


Fig. (a)

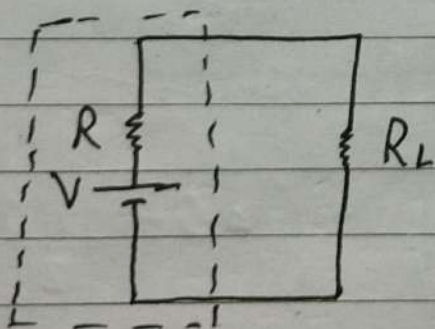


Fig. (b)

$$V = \frac{\pm V_1/R_1 \pm V_2/R_2 \pm \dots \pm V_n/R_n}{1/R_1 + 1/R_2 + \dots + 1/R_n}$$

$$R = \frac{1}{1/R_1 + 1/R_2 + \dots + 1/R_n}$$

Tellegen's Theorem

In any network, total instantaneous power consumed by various element in different branches of network is always equal to zero.

$$\sum_{k=1}^b V_k I_k = 0$$

where, V_k = Branch Voltage

I_k = Branch Current

b = No. of Branches